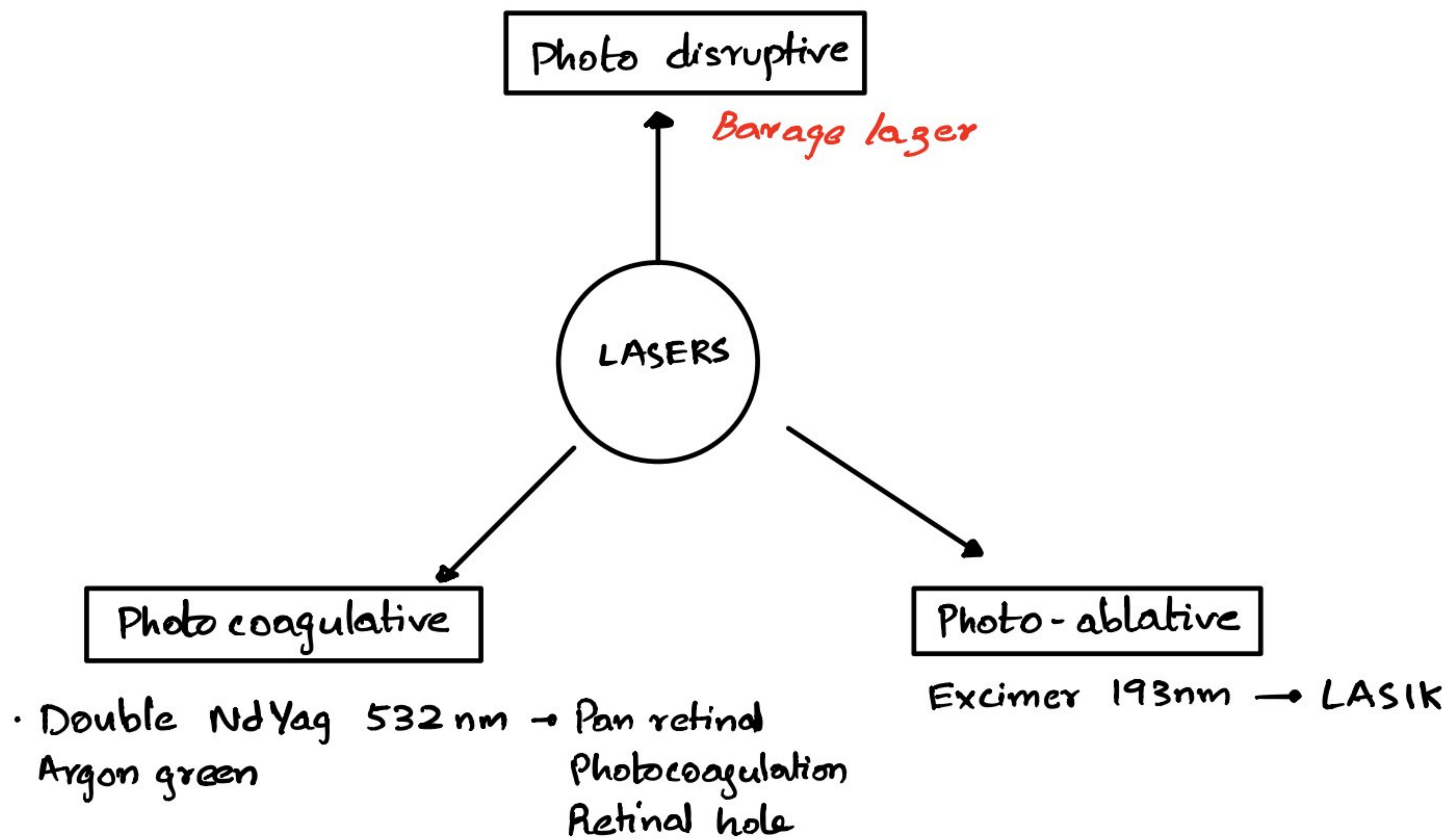
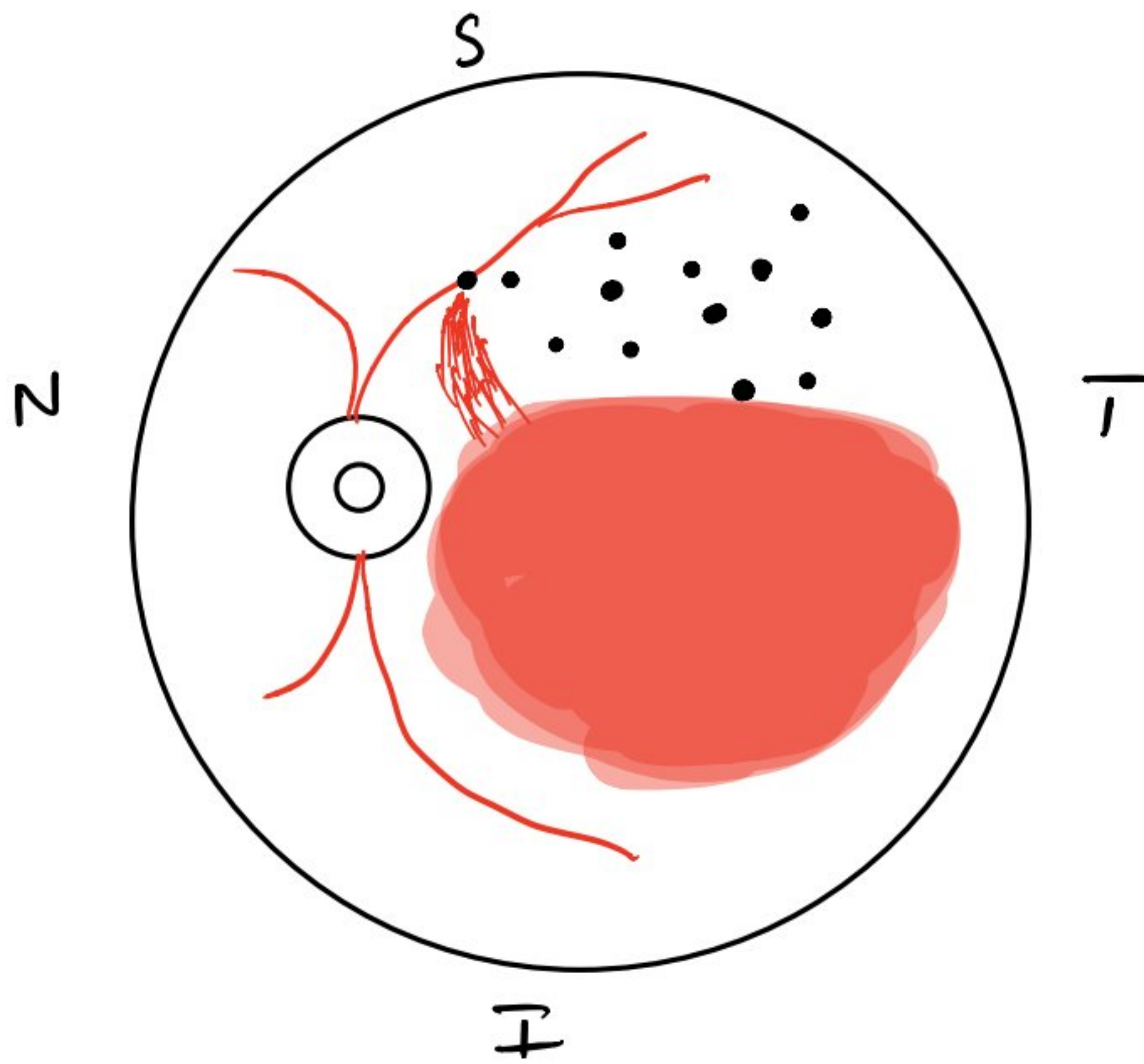


LASERS IN OPHTHALMOLOGY



Q: Which quadrant is lasered first?

sol. Start from inferior quadrant → During laser there will be chance of hitting of vessel if superior quadrant lasered first if accidentally hit any vessel the H'mage settles inferior and we may not be able to visualise the inferior quadrant.

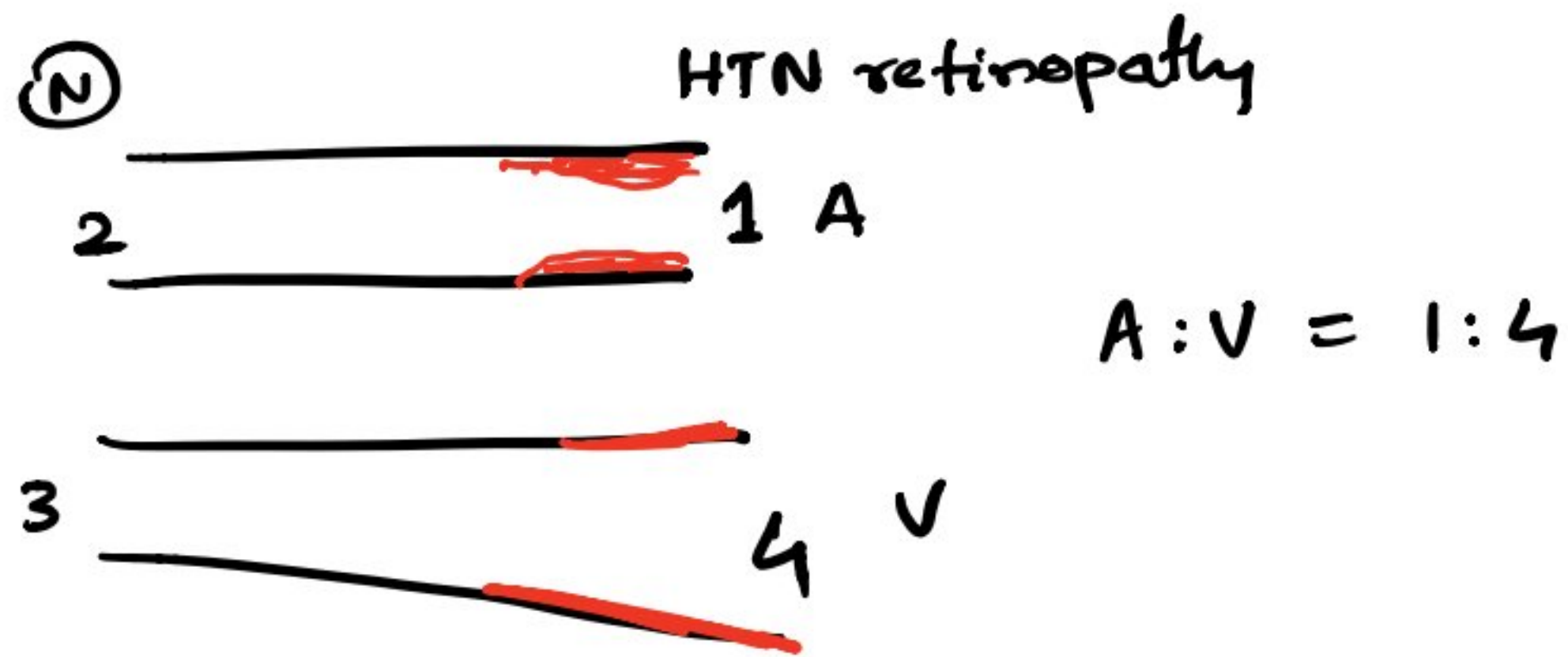


Complication of PDR (Advancement Diabetic Eye d/s)

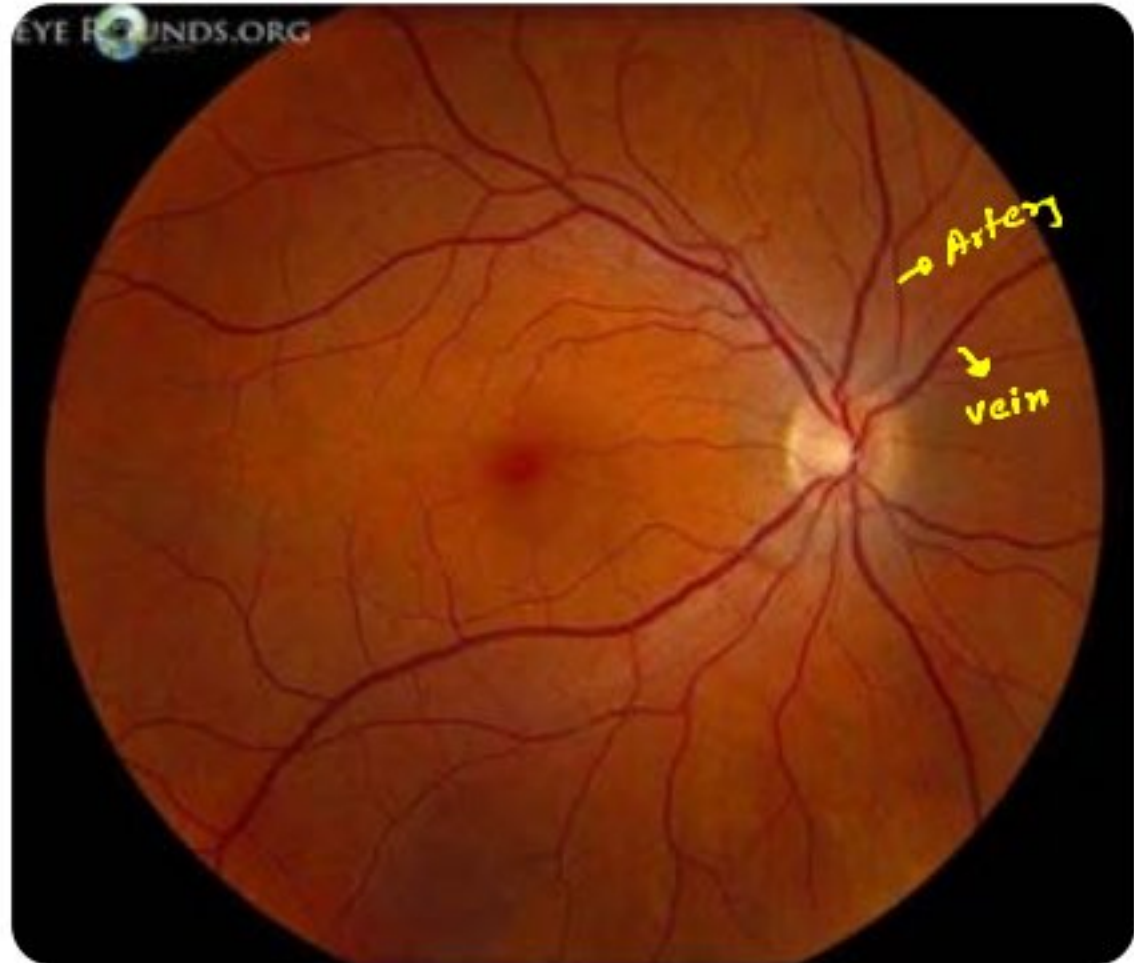
- i) Vitreous H'mage → sudden, painless loss of vision
- ii) Tractional Retinal Detachment → Retina pulled by fibrous new vessels
- iii) NVI → NVA → Neovascular Glaucoma

↳ Rx → Trabeculectomy & Mitomycin - C
or
Aqueous Shunts

HYPERTENSION:



Artery	Vein
-	Thicker
Bright Red	Deeper Red
Shiny streak / reflex	-
Artery crosses above the vein	-



Pathogenesis: In sustained HTN, the inner blood retinal barrier is disrupted and vascular permeability is ↑sed leading to flamed shaped H'mage.

1st sign of HTN retinopathy → ARTERIOLAR GRADING

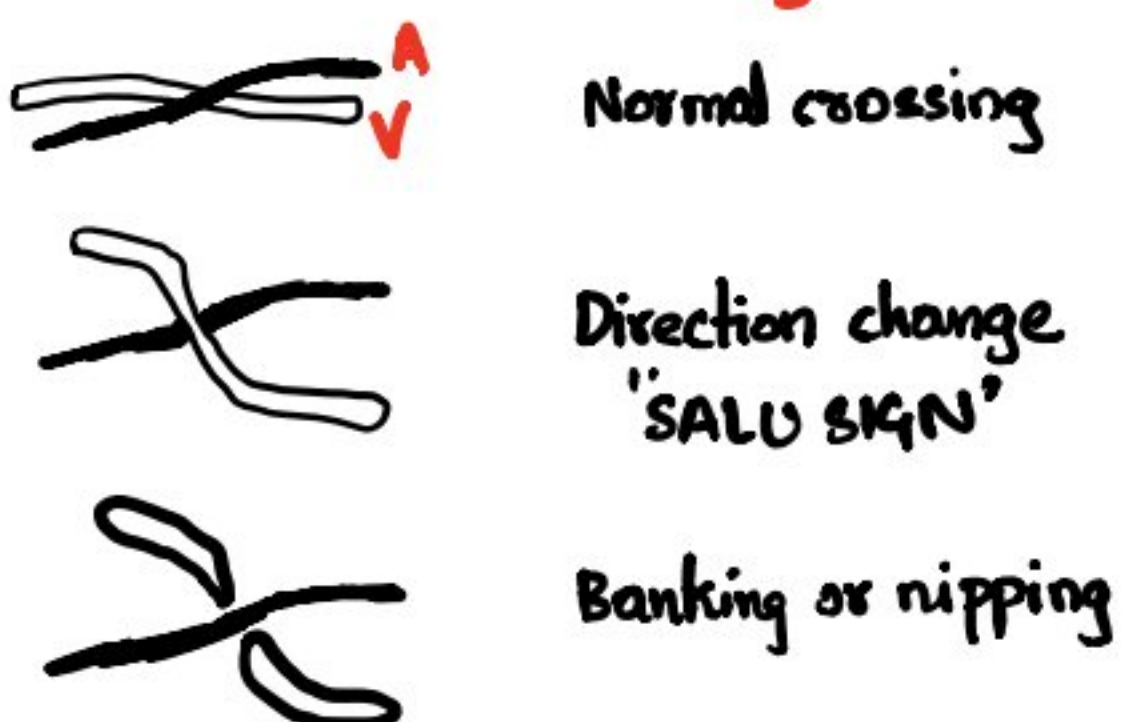
NEET 'P4

Keith Wagner Staging

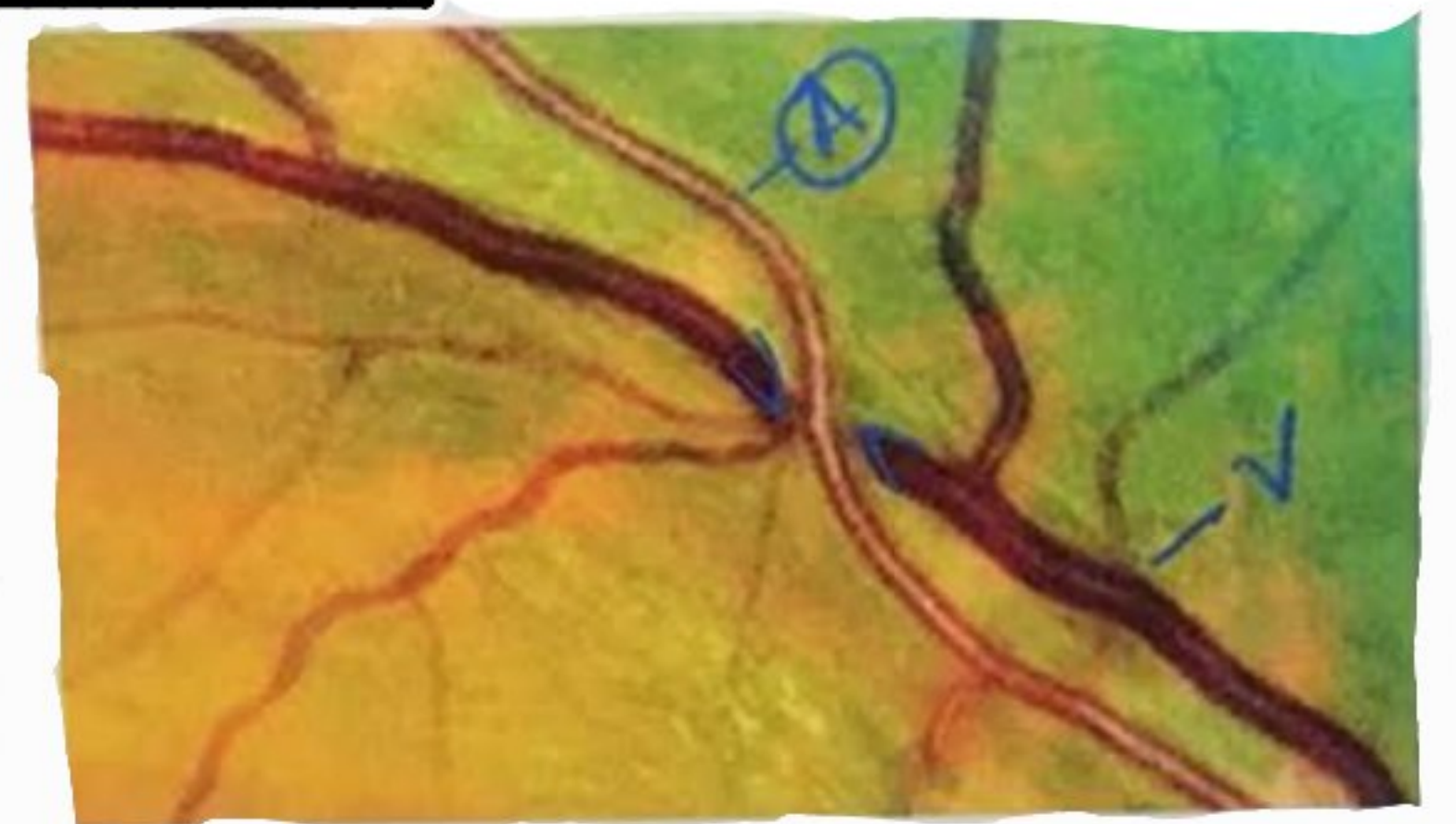
I → Arteriolar attenuation

II → Stiff arterioles compress vein @ Arteriovenous crossing } A-V nipping change

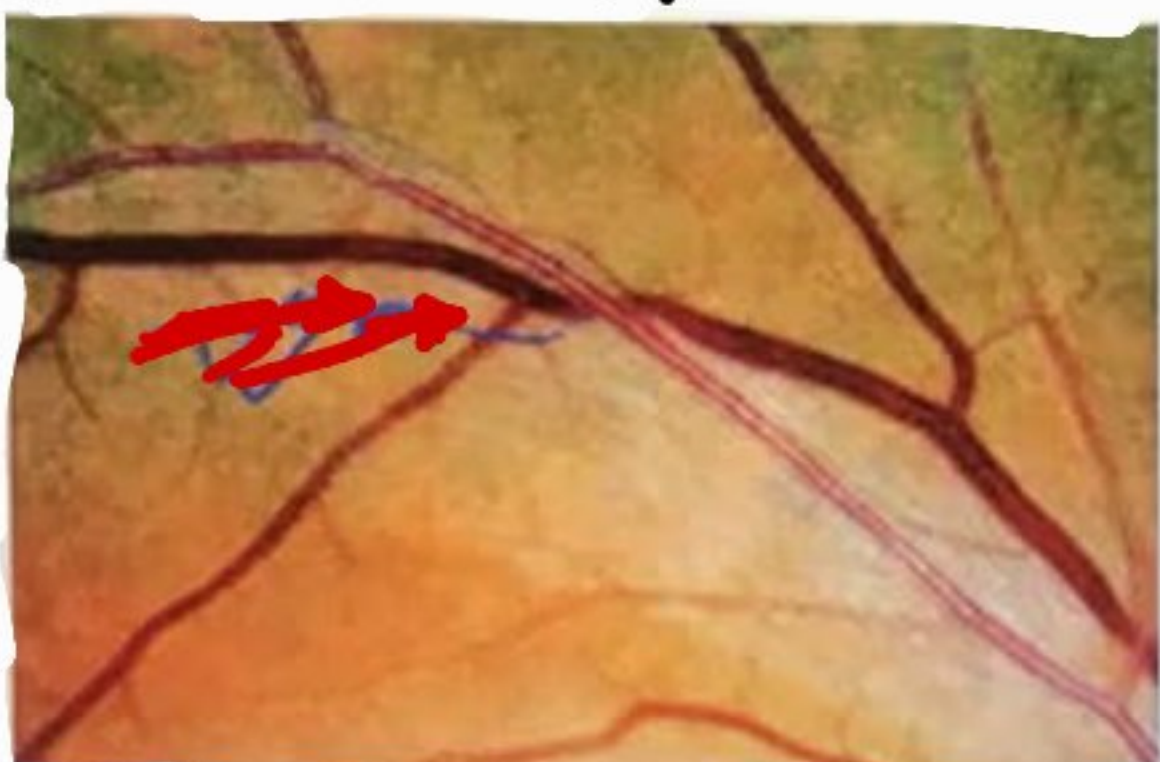
Vessel "Crossings"



GUNN SIGN → Tapering of Vein



BONNETT SIGN



DISTAL BANKING

SALU SIGN → change in direction vein

Grade 3: cotton wool spots (RNFL infarcts)
Copper wiring of arteriole



Copper wiring of arterioles

Grade 4 → Silver wiring of arterioles



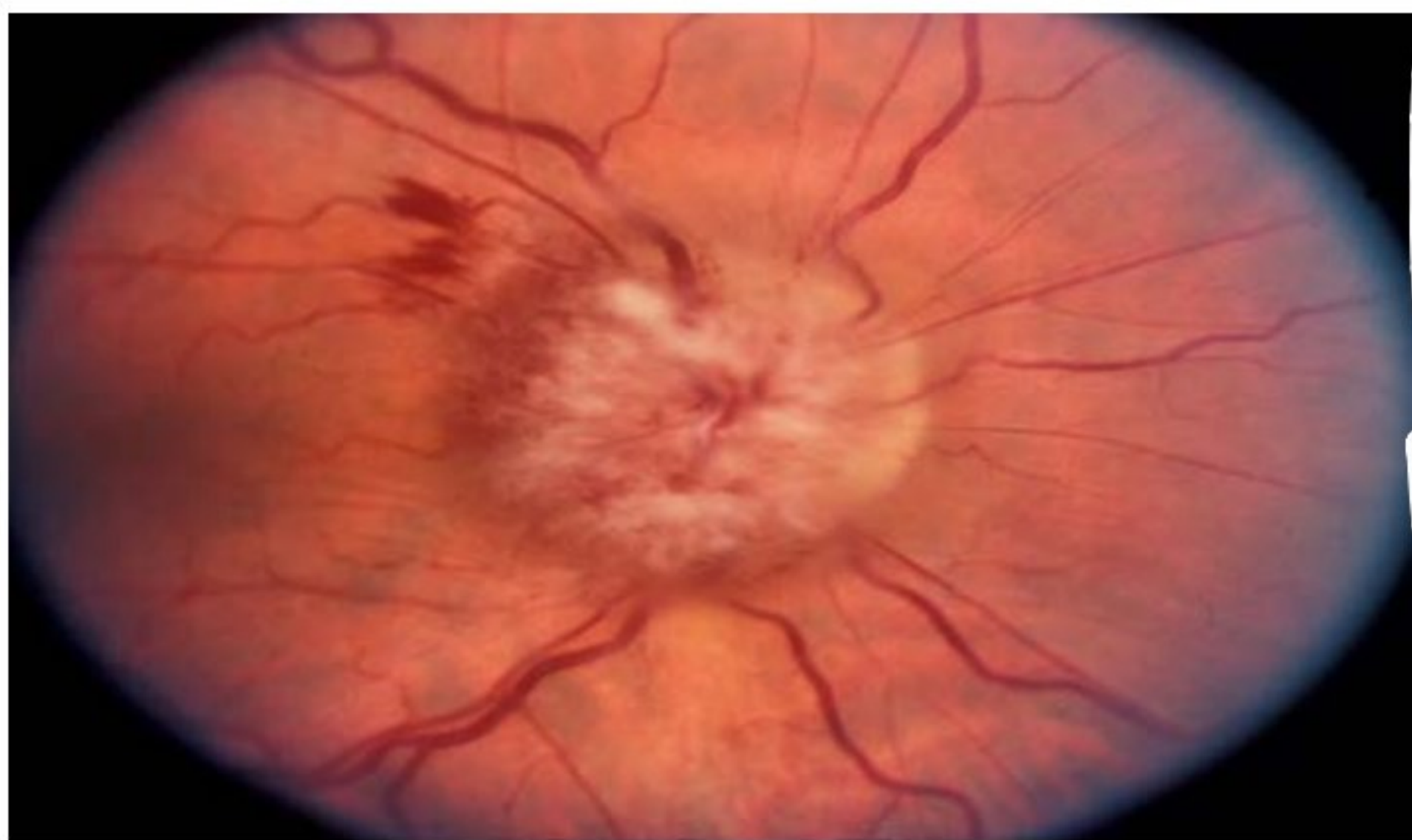
Silver wiring of arterioles

Other changes

i) Macular star → Hard exudates arranged in form of a star in macula (Henle's loop)

ii) Papilledema → B/L disc edema

↳ **Malignant HTN**
Hallmark



Other D/D of Macular Star

- 1) Grade 4 HTN Retinopathy
- 2) PIH (pregn. induced HTN) Retinopathy
- 3) Neuroretinitis

Mx → Rx of the HTN

Pregnancy Induced HTN Retinopathy:

- Hallmark : B/L serous RD d/t breakdown of outer Retinal barrier → subretinal fluid.
- Earliest sign → Nasal arteriolar spasm
Macular Stars

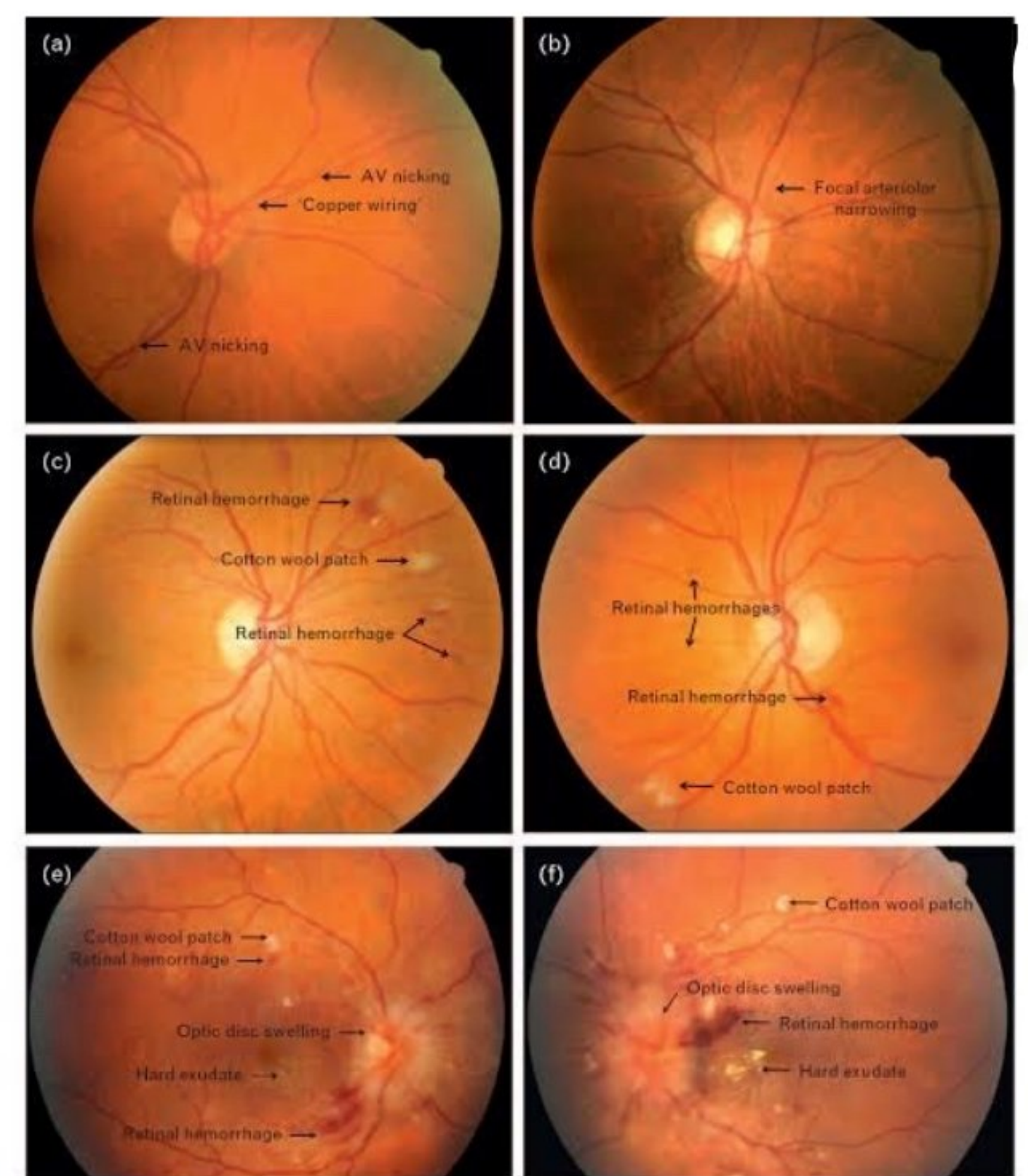


Fig. 10.10 Retinal fundus photographs of mild (a, b), moderate (c, d), and malignant (e, f) hypertensive retinopathy, as per Keith-Waggoner-Barker classification.

FINDINGS	LAYER
1) Microaneurysm	INL
2) Dot - Blot H'mage	OPL
3) Hard exudates	
4) Macular Star	Henle's layer
5) Cotton wool spot (soft exudates)	RNFL
6) Flame shaped H'mage	

Q: The Mizuo phenomenon is s/i

- ✓ a) Oguchi's disease
- b) Fundus albipunctatus
- c) Fundus Flavimaculatus
- d) Retinitis Pigmentosa

MIZUO PHENOMENON is s/i Oguchi's d/s.

Oguchi's d/s $\Rightarrow$ congenital stationary night blindness with Ab^(N) fundus.

AIIMS' 10
AIIMS' 11

Q: Exudative Retinopathy in HTN is d/t

- a) Coarctation of aorta
- b) Type 1 Takayasu's arteritis
- ✓ c) Renal artery stenosis
- d) Diabetic atherosclerosis

Retina

Dystrophy

- Heritable
- Usually B/L

1) Rods > Cones dystrophy $\rightarrow$ RETINITIS PIGMENTOSA

- (MC) hereditary Retinal dystrophy
- young pt.
- Night blindness

Inheritance of RP

- i) AR $\rightarrow$ (MC)
- ii) AD $\rightarrow$ Best Px
- iii) XLR $\rightarrow$ Worst Px
- Degeneration of RODS $\xrightarrow{f/b}$ cones

{Inverse RP}: Degeneration of cones $\xrightarrow{f/b}$ Rods

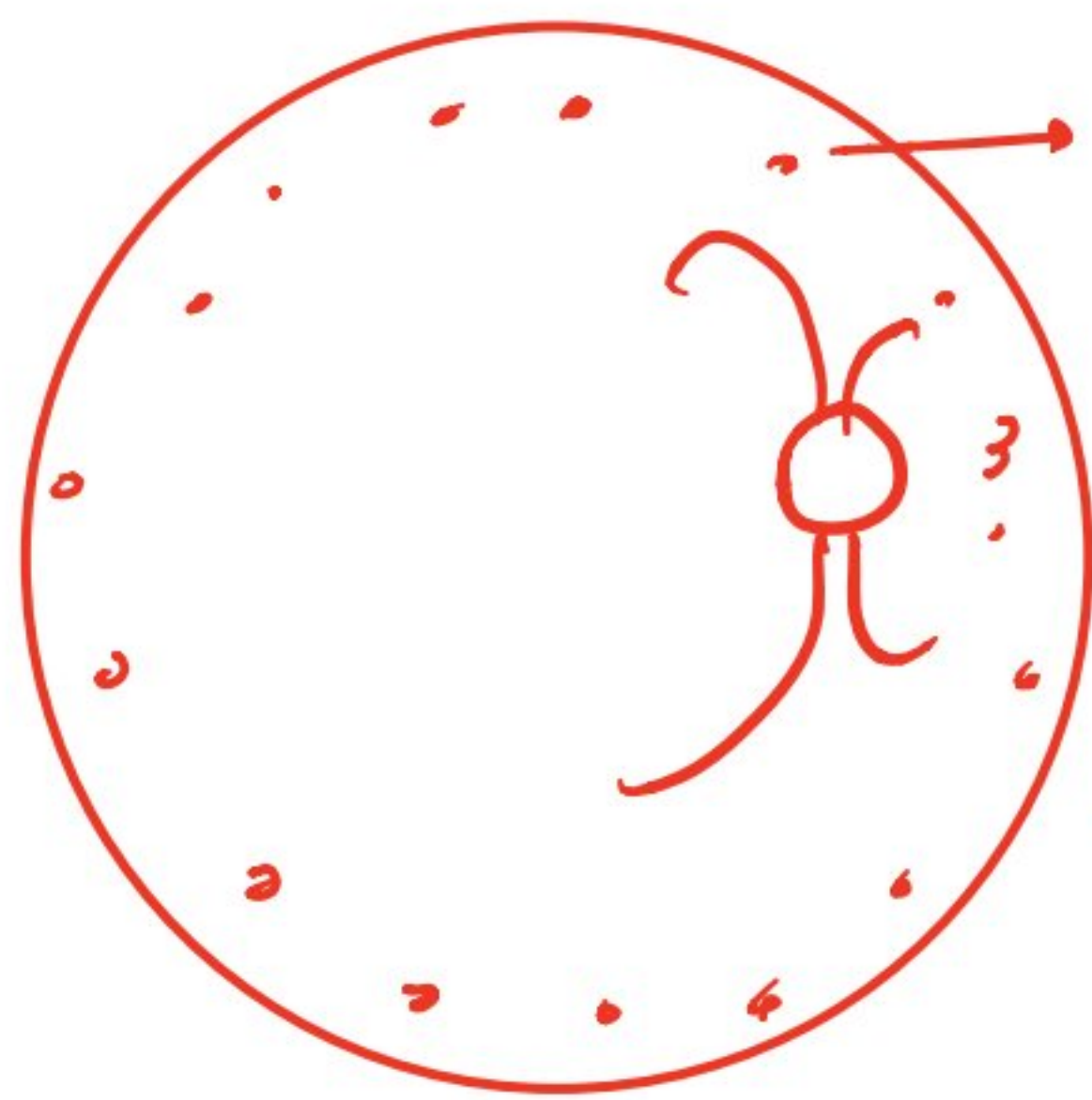
Degeneration

U/L

Acquired

B/L

Rods
↓
RHODOPSIN
req. Vit A.
gene = 301



Rods are max-density

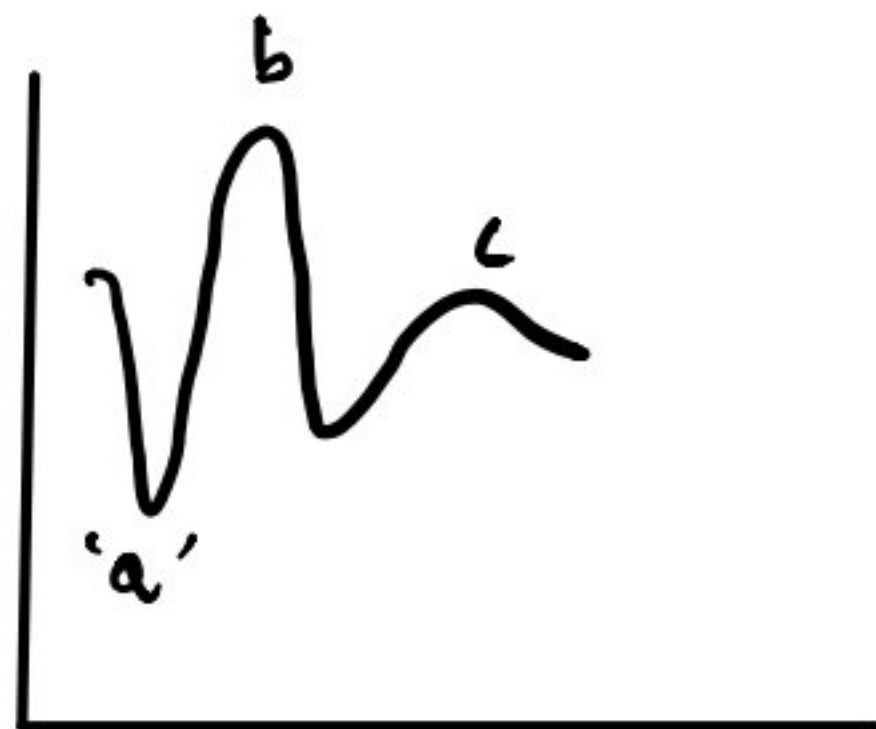
@ equatorial region
(mid-peripheral retina)

SCOTOPIC VISION : Vision in dim light

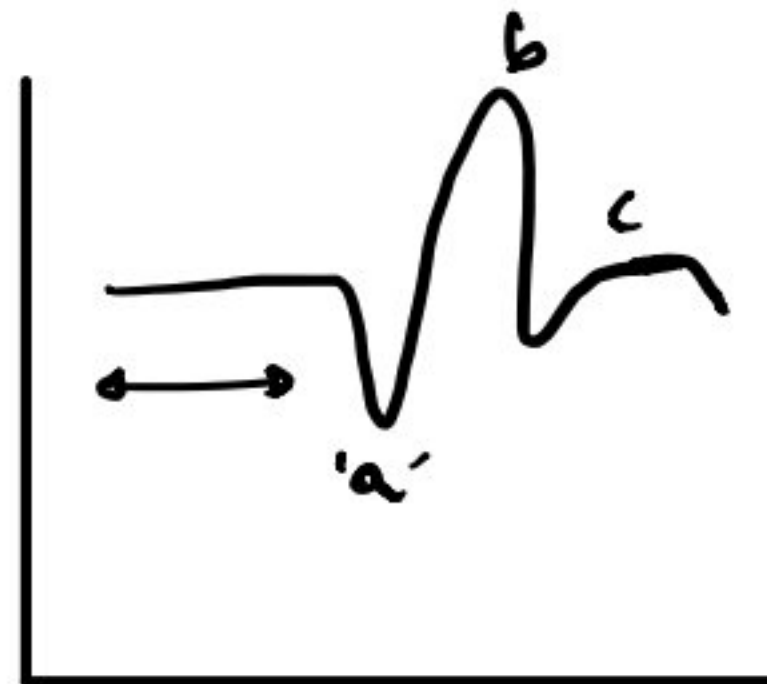
1st Symptom → NYCTALOPIA {night blindness}

1st sign → on ERG : Delay in 'a' wave

Increase in implicit time of 'a' wave



NORMAL



RP.

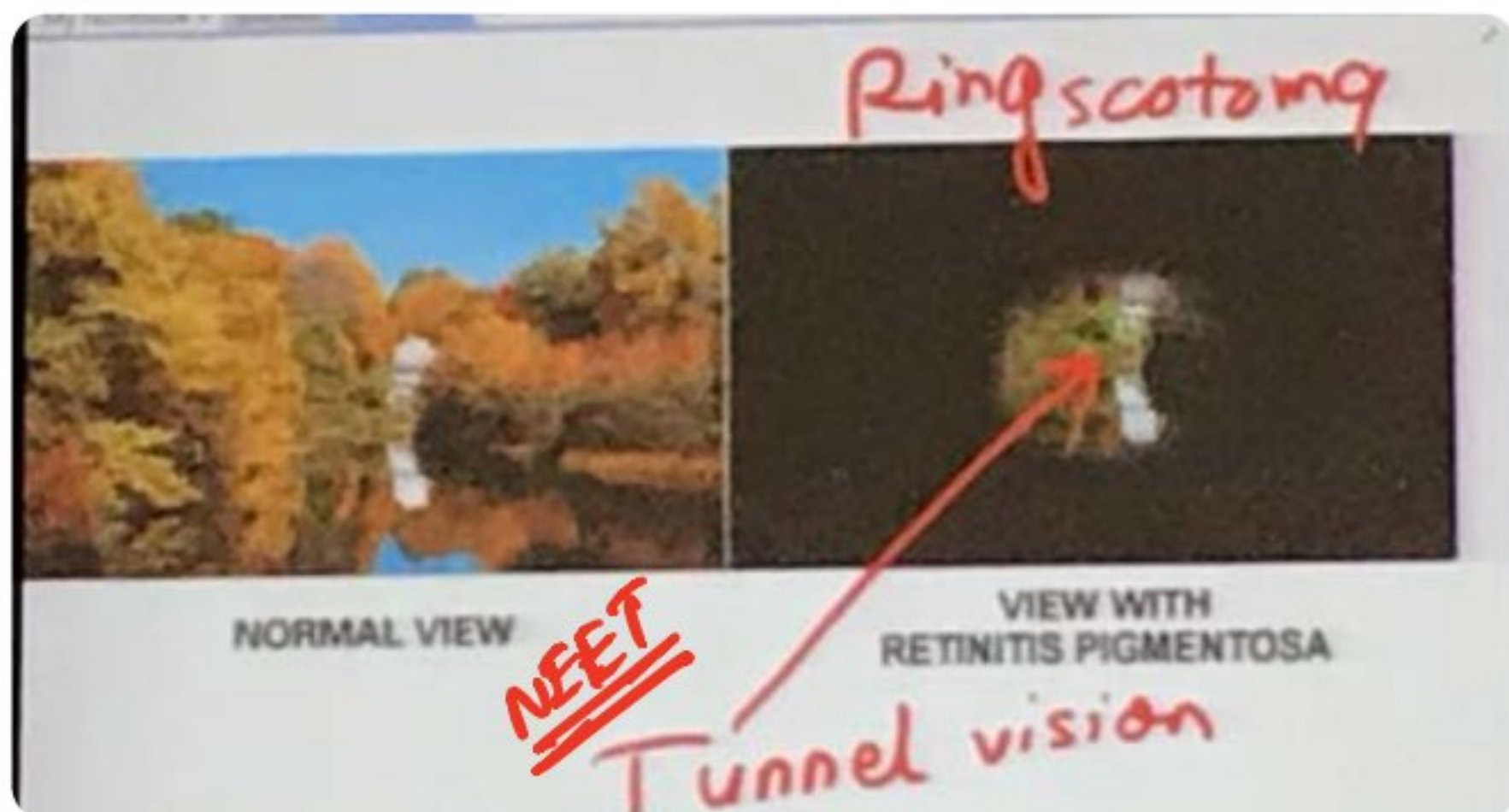
→ Delayed dark adaptation time

ERG findings in RP

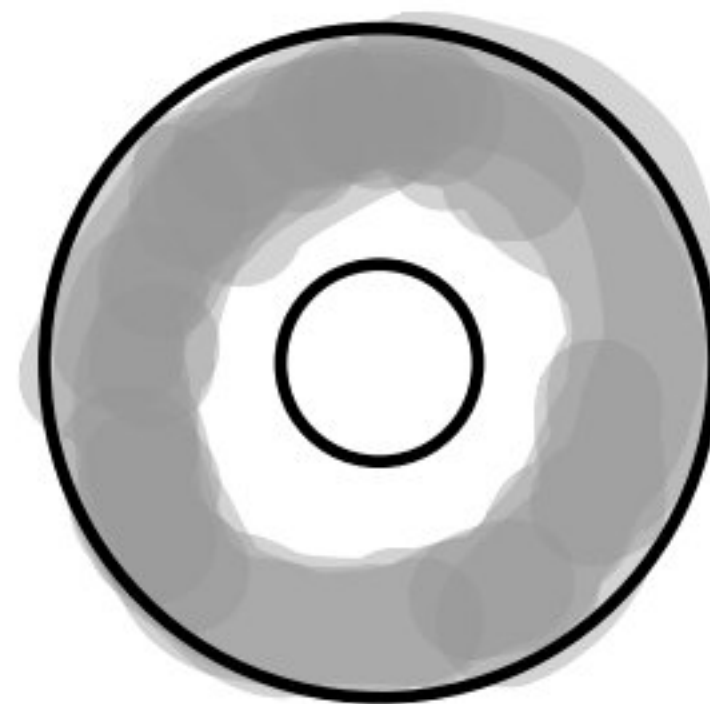
1, 'a' wave is delayed
↓ later

2, 'a' wave is Absent

Advanced RP { Extinguished ERG
or FLAT ERG



- Ring scotoma → Tunnel vision
↳ on Perimetry (visual field)
- Double Ring Scotoma → RP + macular edema



Fundoscopy Triad:

- 1, Arteriolar Attenuation
- 2, Waxy disc pallor of optic disc
- 3, Bony specules { pigmentation in periphery }



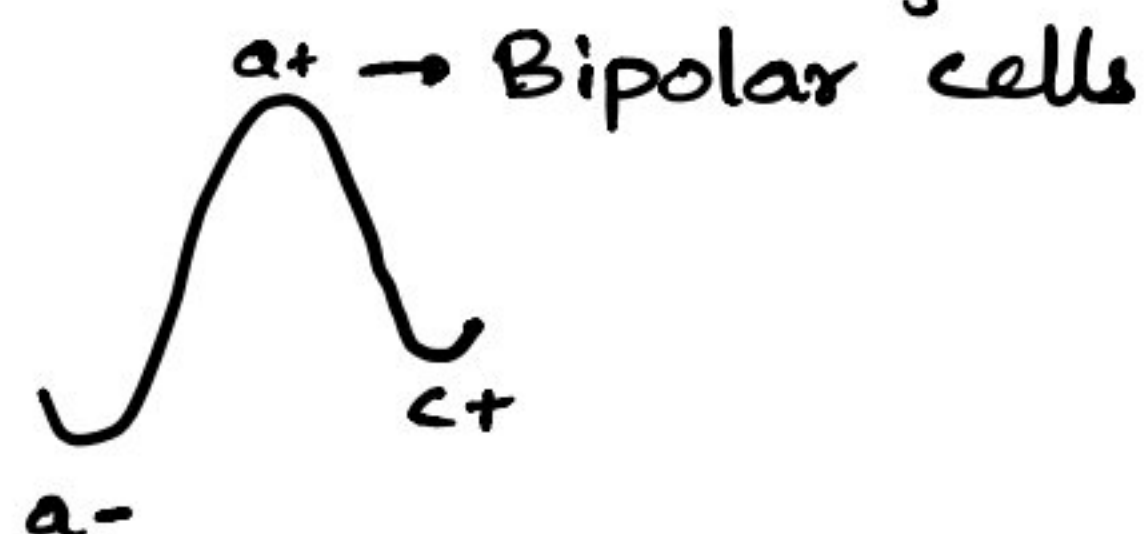
AIIMS PGx2

Q: A patient presented with peripheral field loss and fundus examination shows waxy exudates & pale disc with pigmentation around retinal vessels. Dx?

- Behcet's Sx
- Chorioretinitis
- Open angle glaucoma
- Retinitis Pigmentosa**

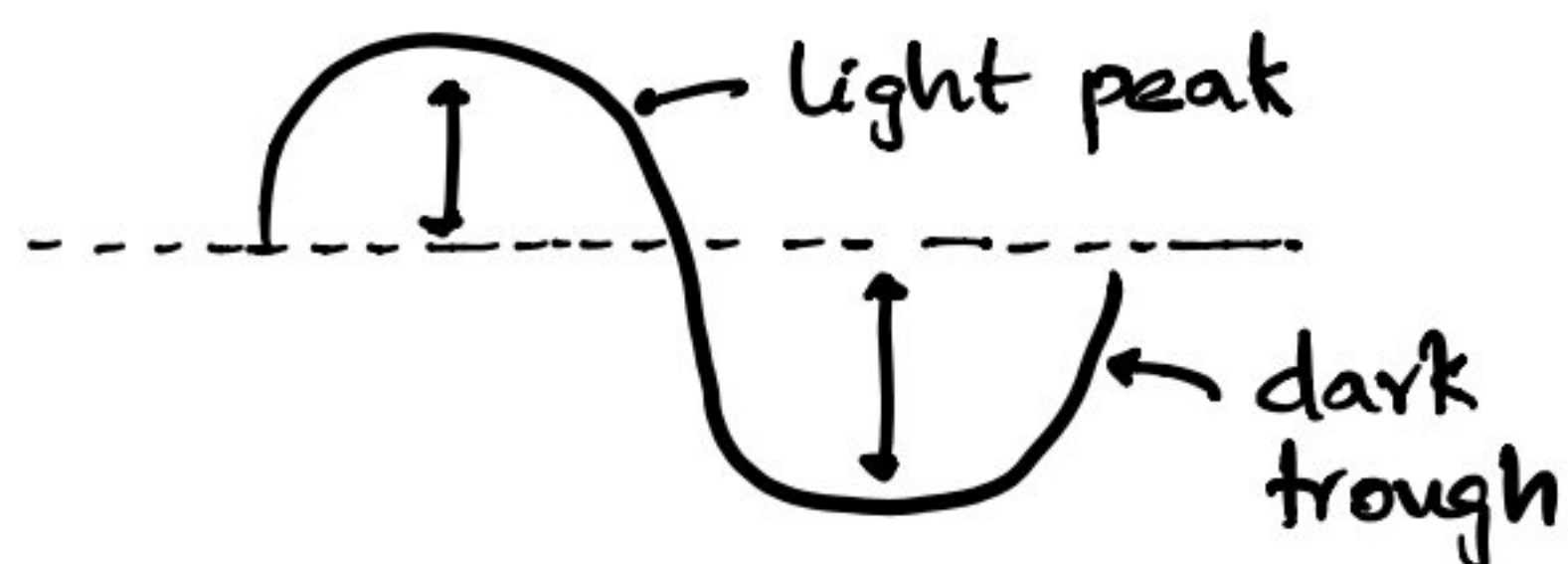
Ans:

ERG electroretinogram



- a wave reduced first @ Rods mainly
- ERG $\Rightarrow$ scotopic > Photopic (Rods) (cones)

2, EOG $\rightarrow$ Electrooculogram & assess RPE



Arden ration = $(N) > 1.85$

Ab $(N) < 1.85$

Flat < 1.25

subnormal < 1.50 curve

= ERG $\downarrow$
early stage
EOG $(N) > 1.85$
 $\downarrow$
BEST d/s opposite

OTHER EYE finding in RP

- 2° Keratoconus
- Primary Open angle glaucoma (POAG)
- Posterior subcapsular cataract (PSC) $\rightarrow$ B/L [complicated cataract]
- Macular edema



Macular Edema

Rx: No treatment of dystrophies supplement $\rightarrow$ with Vit. A

DOCOSAHEXANOIC ACID (w/ 3 fatty acid)

These are deficient

C → Cockayne → SNHL ⊕

R → Refsum → ATAXIA ⊕

U → Usher (MC) → SNHL ⊕

S → Kearns Sayre → Heart Blockers + CPEO

H → Halgren

B → Bassen Kornzweig → A-B Lipoproteinemia

B → Bardet - Biedl syndrome
Laurence - Moon

- Polydactyly
- Obesity
- MR
- Hypogonadism

- steatorrhea
- Defective absorption of fat soluble vitamin
- Peripheral blood smear ⇒ Prickle cells / acanthocytes

Acanthocytosis.

KEARNS - SAYRE Sx → • Mitochondriopathy & Mitochondrial inheritance
• Cardiac conduction defects
• CPEO [chronic progressive external ophthalm.]
↳ EOM + LPS paralyzed [FROZEN EYEBALL + ptosis]



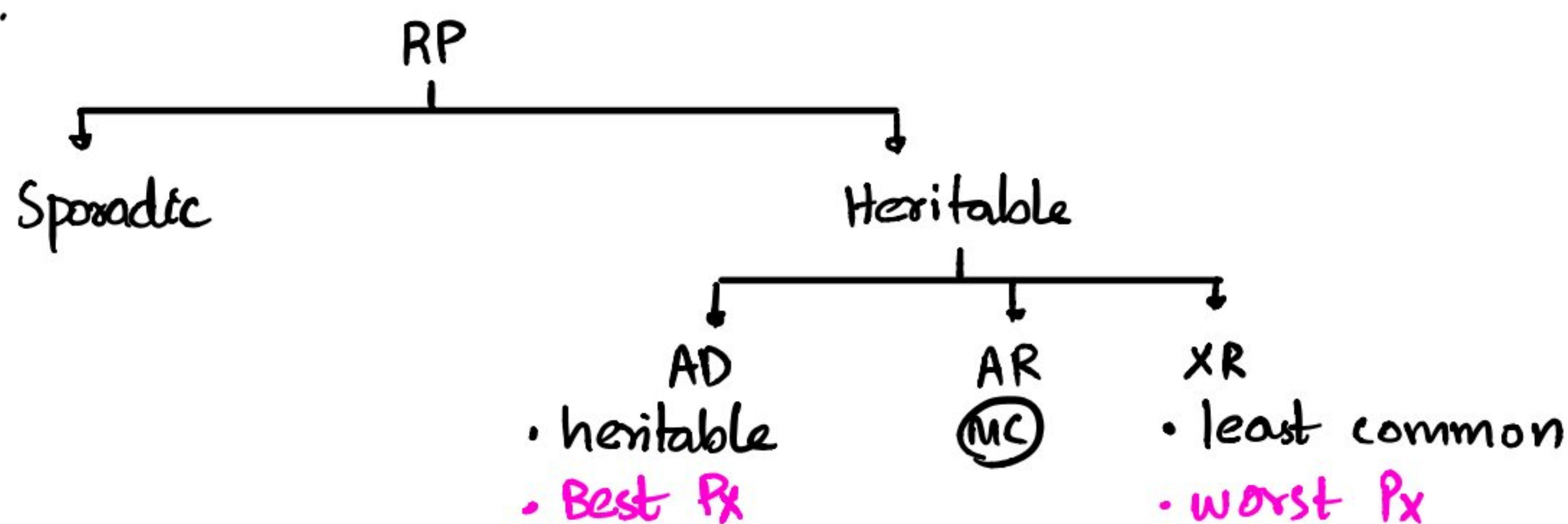
a. Fundus examination → Pigmentary retinopathy & restriction of all eye movement
↳ B/L drooping of lids. Inx to be done?

sol. a) ERG b) EMG c) EEG d) ECG

NARP Syndrome :: N → peripheral neuropathy
A → Ataxia
RP → Retinitis pigmentosa

Refsum d/s → • Phytanic acid oxidase Absent
• RP + Ataxia

Types ::



PHOTORECEPTORS:

Cones

Red > Green > Blue

(max.)

(min.)

↓
(chrom. X
(XLR))

↓
Chrom. 7
(AR)

Rod: Cones

20 : 1

120 million rods

6 million cone → max^m density @ foveola



Red colour defect → **PROTAN**

Green colour defect → **DEUTERAN**

Blue colour defect → **TRITAN**

Testing Colour Vision:

1) ISHIHARA'S Pseudoisochromatic charts → only for Red-green defects not for TRITAN anomaly

2) H. RR chart
3) City university chart } for TRITAN anomaly

Gold standard → **NAGEL ANOMALOSCOPE** {color mixing}

• Colour vision d/t occipital lobe **area V₄**

• **ACHROMATOPSIA** → lesion of area V₄

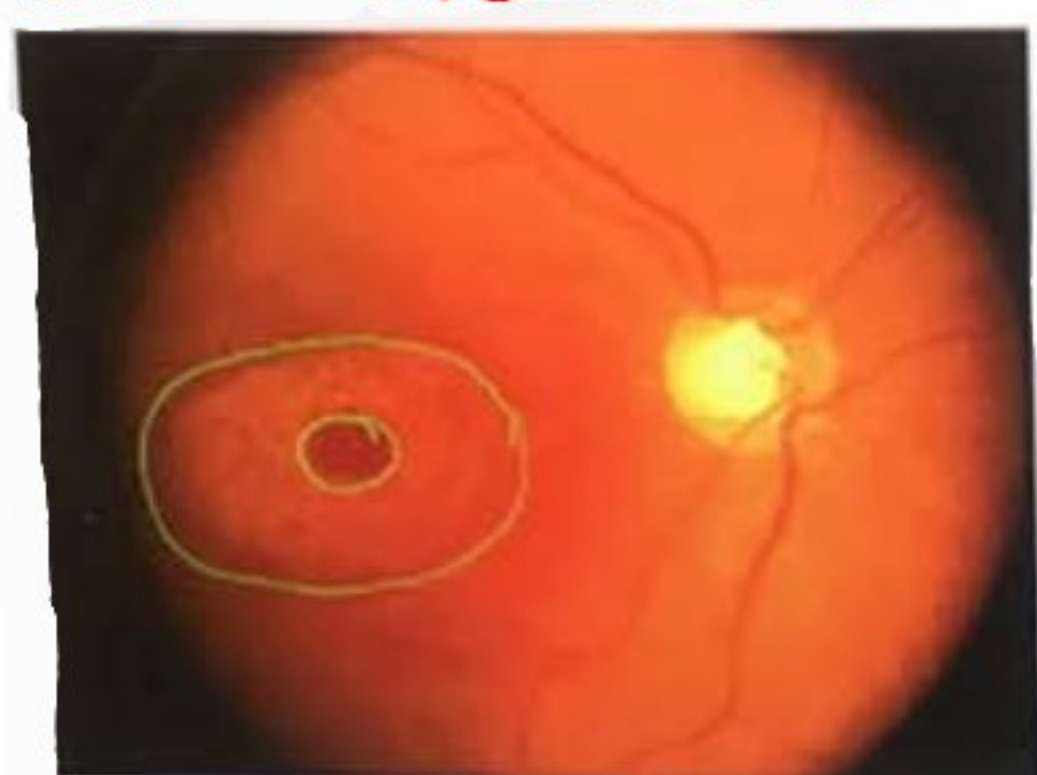
• Photopic vision → Vision in bright light
Cones responsible.

Congenital cone deficiency:

• Day blindness → **HAMARLOPIA**

• Bulls eye maculopathy

BULL'S EYE MACULA → i) Congenital defb. of CONES



ii) Chloroquine Toxicity
iii) Hydroxy chloroquine

MACULA FUNCTION TEST

- 1) Visual Acuity → 6/6
- 2) Colour Vision
- 3) 2- Point discrimination test
- 4) Maddox Rod test
- 5) Arnolds Grid → for metamorphopsia
- 6) Photostress test

Macular disorders

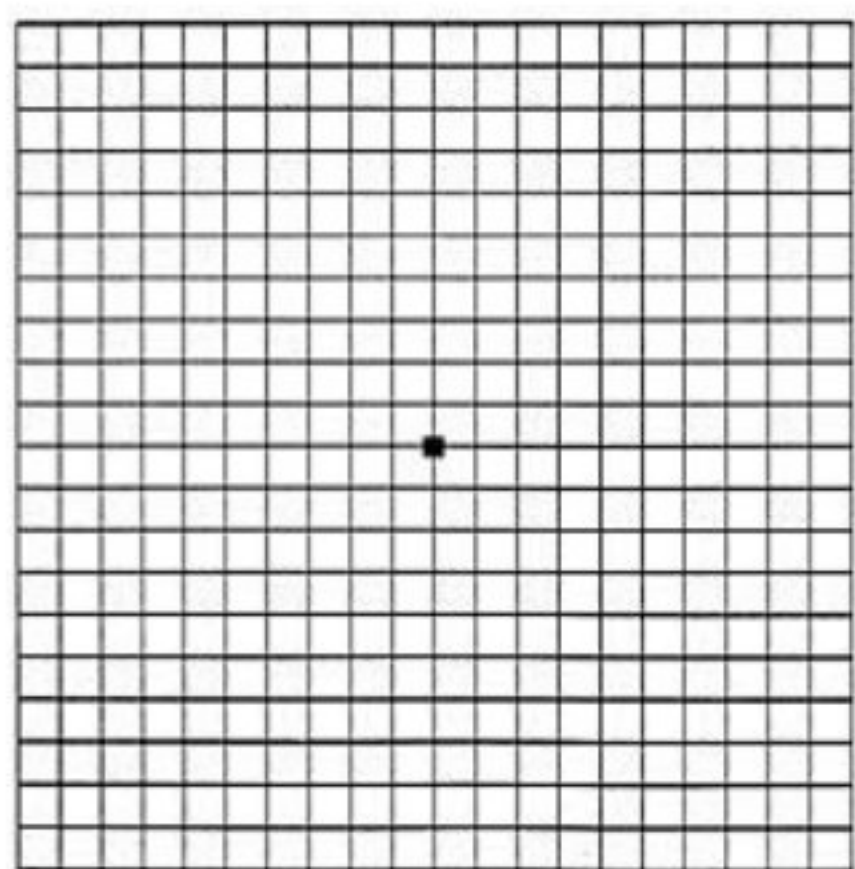
- Vision $\neq$ 6/6
- Vision will not improve with pinhole.

Symptoms:-

1) METAMORPHOSIA



2) Central Scotoma



NORMAL

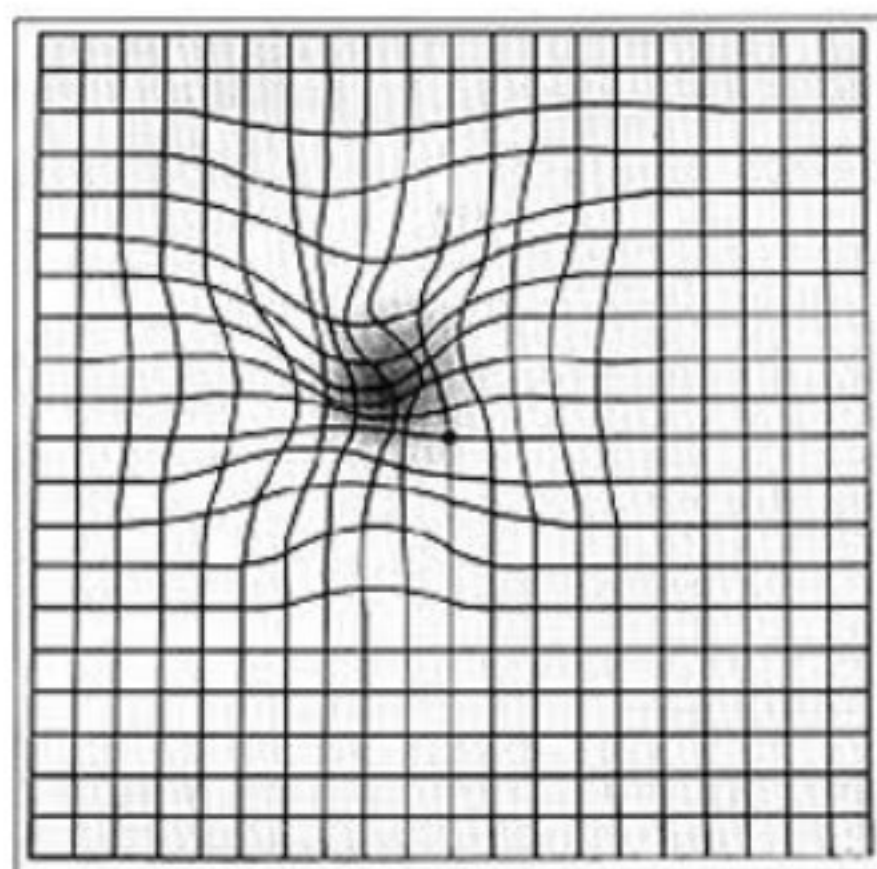
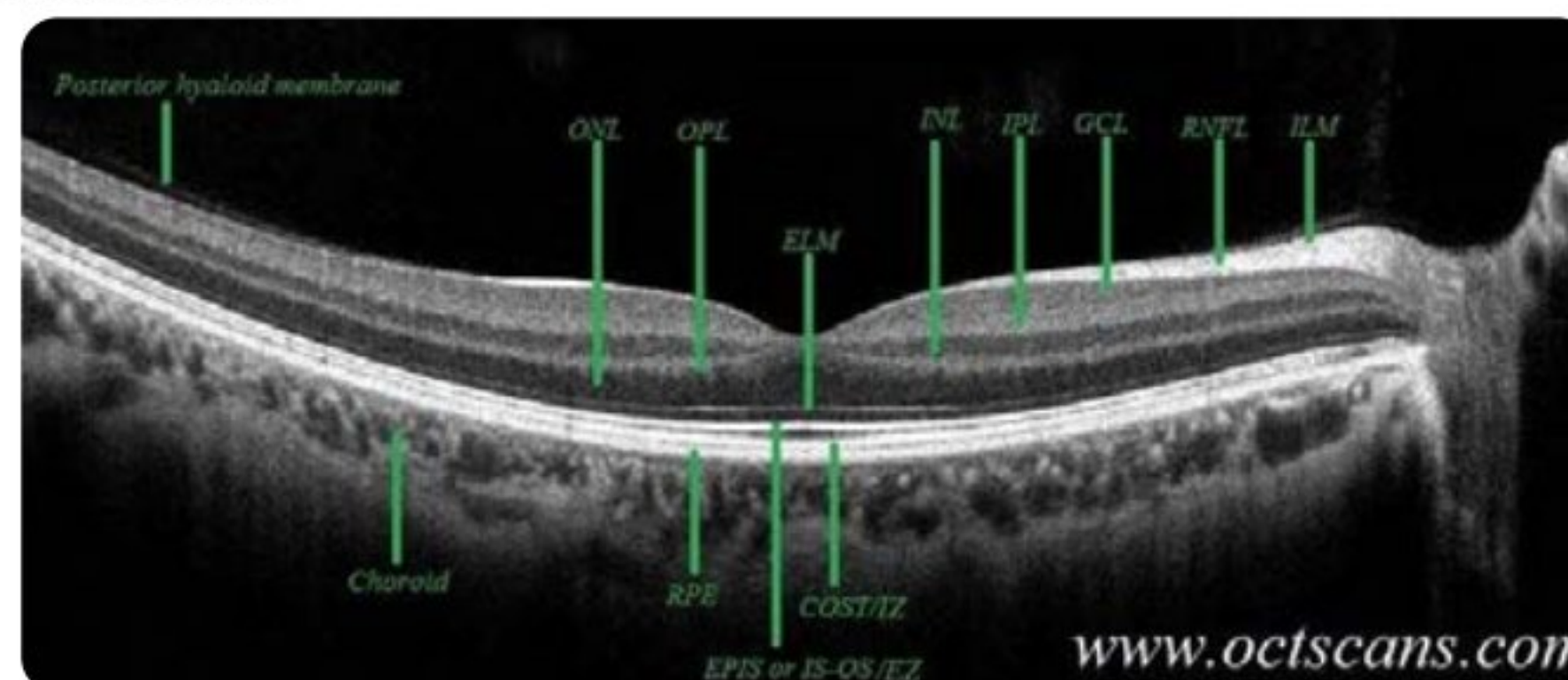


Image courtesy of Nat. Eye Institute, Nat. Institutes of Health

Amslers Grid

① Macular thickness } OCT measured



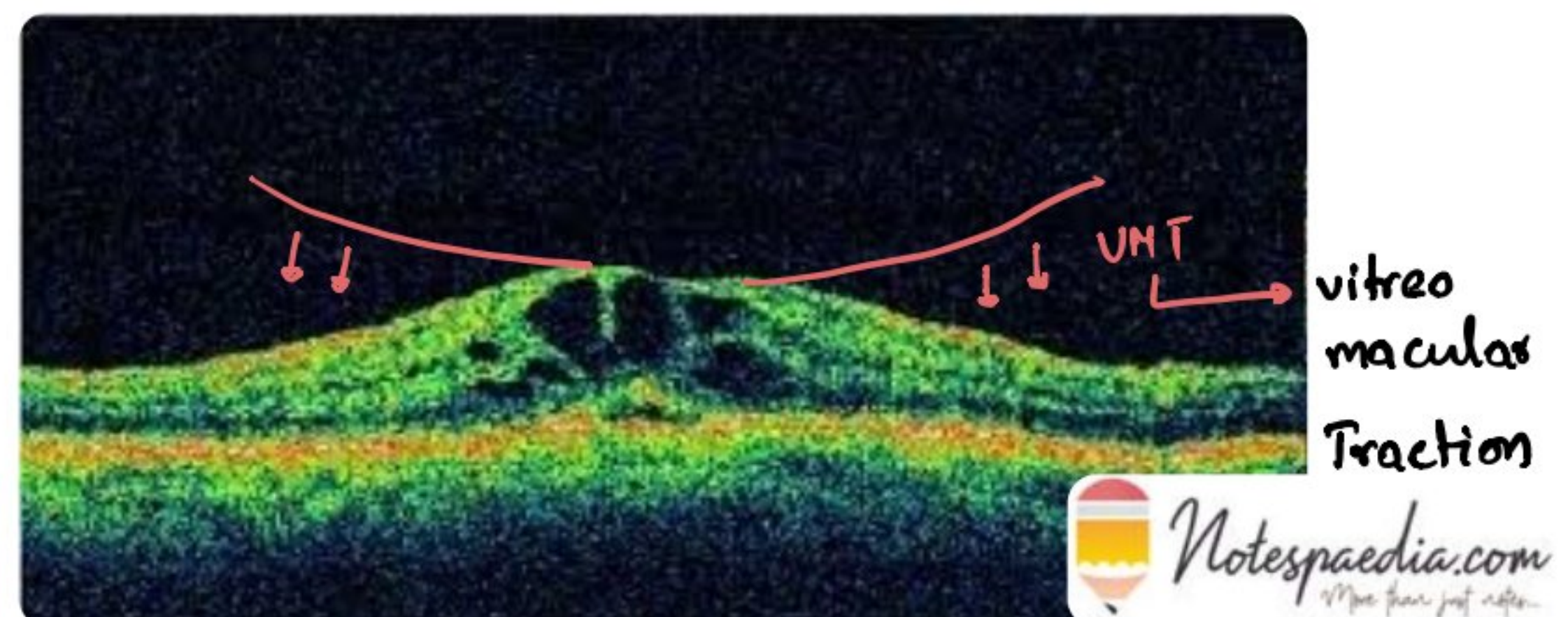
Ab(N) OCT:-

1) Cystoid Macular Edema

cause: i) DM → NPDR

ii) CRVO

iii) Cataract Sx post Irwin - Gass Sx



- iv) Internal Uveitis
- v) Posterior Uveitis

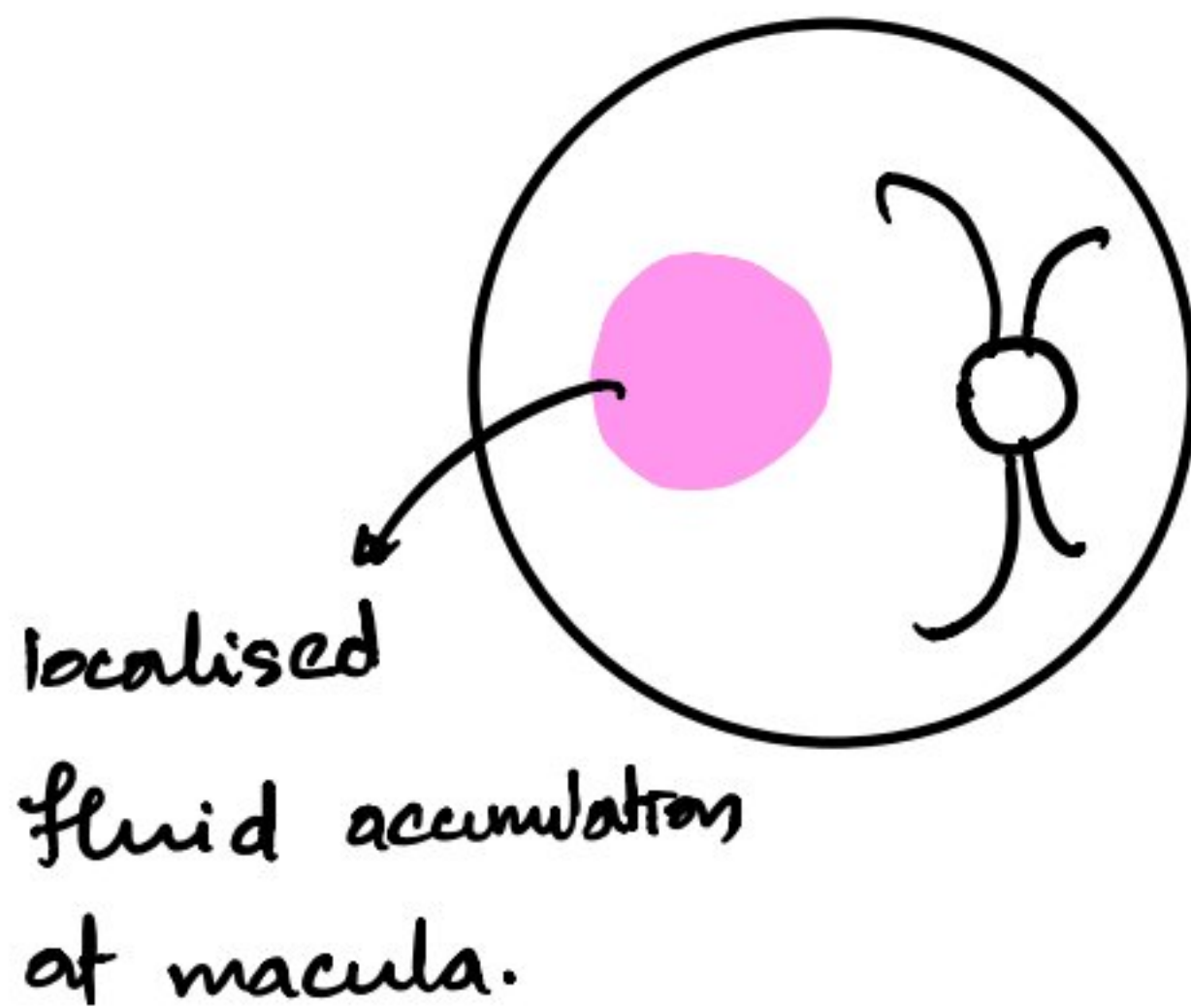
- Cyst edema at outer plexiform layer / Henle's layer
- Honey comb appearance → On ophthalmoscopy
- Fluorescein angiography → Flower petal appearance on FFA { due to break or blood retinal barrier }

Mx: **Observation**

- IV injection of steroids
- IV injection of anti-VEGF
- Pars plana vitrectomy (Refractory CME → VMT)

ii) CENTRAL SEROUS RETINOPATHY / CS Choroidopathy

- d/t Chorocapillary Hyperpermeability → Spontaneous serous detachment of neurosensory retina in macular region d/t fluid accumulation in the subretinal space { B/w RPE & neurosensory layer }



Symptoms

- i) Sudden painless loss of vision
- ii) Metamorphopsia
- iii) Central **Positive** Scotoma
- iv) Micropsia & Edema → Photoreceptor separated from each other }

R.F.:

- 1) Emotional Stress → **Type A personality**
- 2) Endogenous Hypercortisolism
- 3) Exogenous steroids

Inx:

Ophthalmoscopy → serous RD ⇒ Boundary as Ring Reflex }

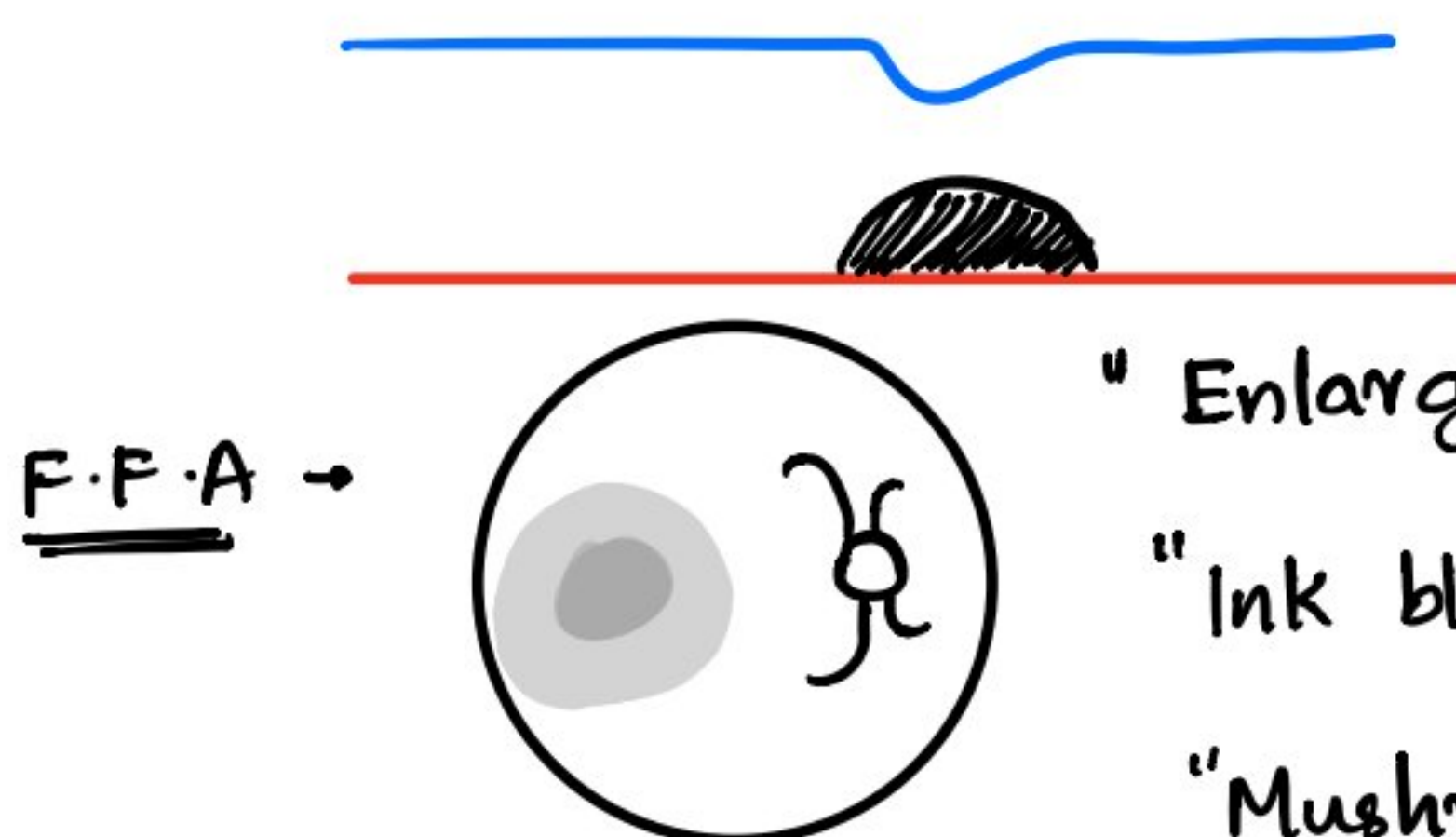
Q:
FOVEAL DETACHMENT



Central serous retinopathy with foveal detachment

Foveal Reflex Absent

InxOC → OCT



Management :: Stop offending drug + Reassurance
 • Laser photocoagulation (Focal) → Spontaneous resolution.

RETINAL DYSTROPHIES contd.

2) BEST d/s / Vitelliform Macular Dystrophy

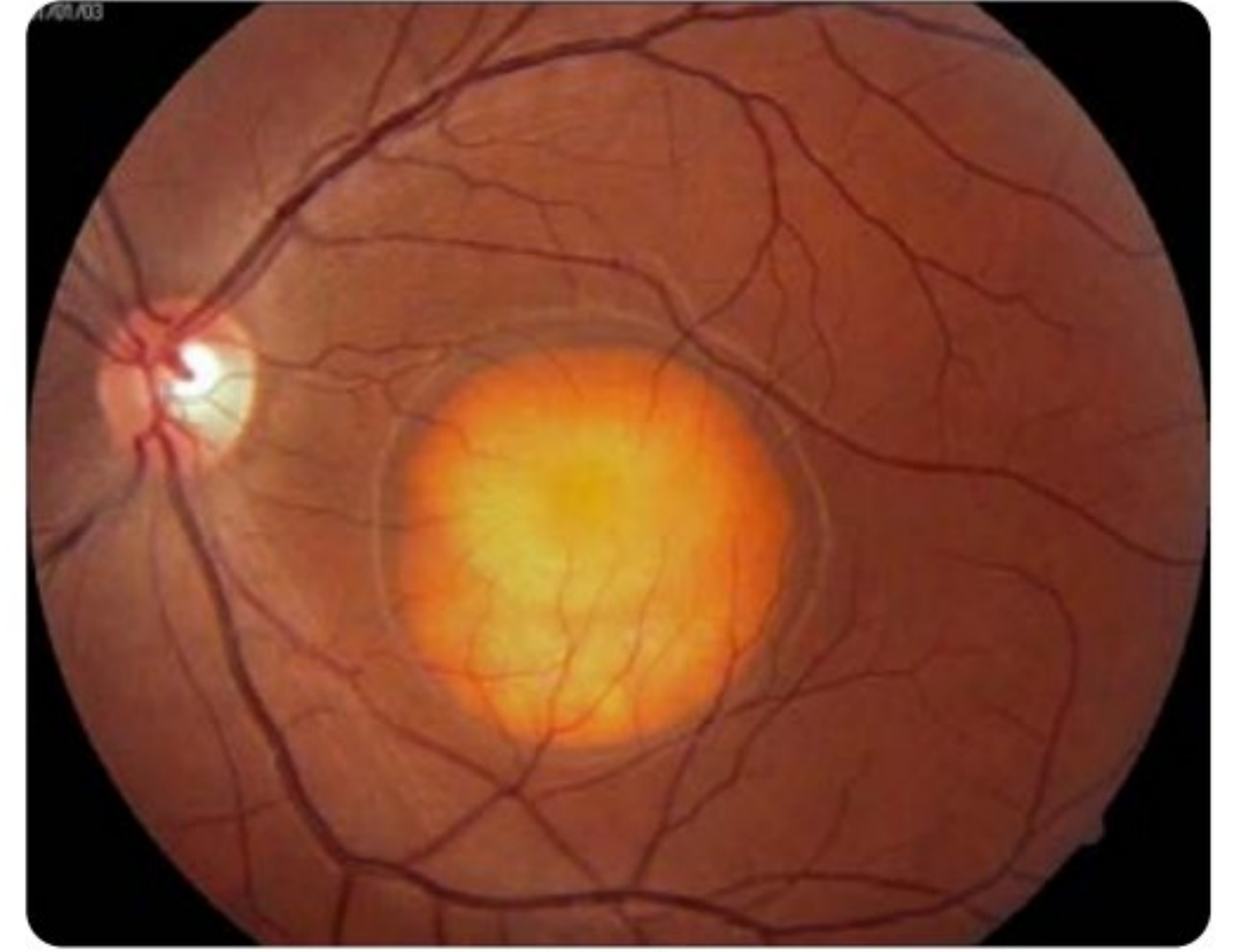
Mnemonic
 B → AD

- affect RPE
- AD, chrom 11 → mutation of VMD 2 gene
- Egg yolk stage d/t Lipofuscin deposit.

No vision loss

- scrambled egg → Pseudo hypopyon

vision ↓



BEST Disease

Inx :: Early stage

Late stage

- | | | |
|----------------|---|-------|
| • ERG (N) | → | ERG ↓ |
| • EOG ↓ < 1.85 | | EOG ↓ |

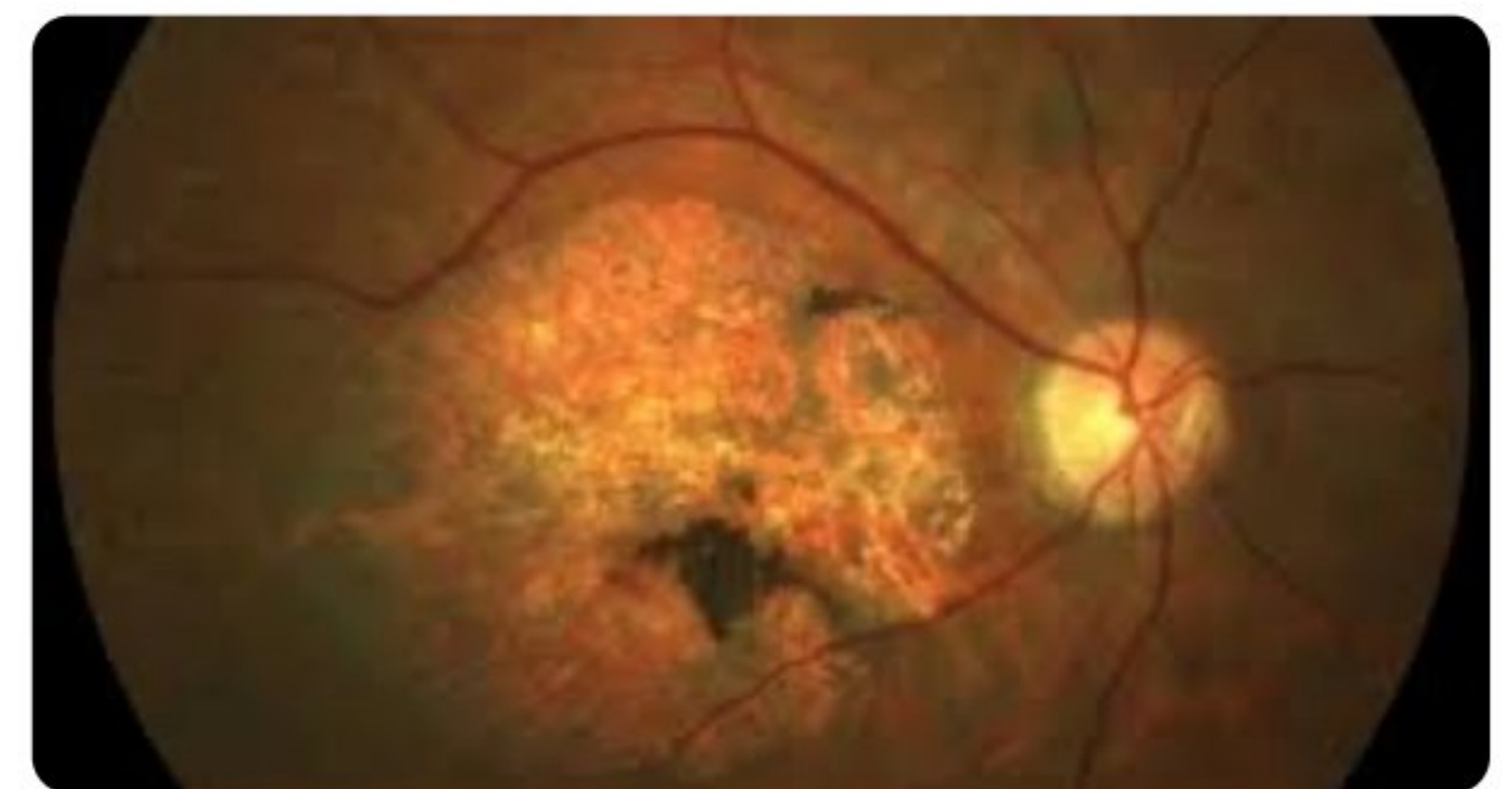
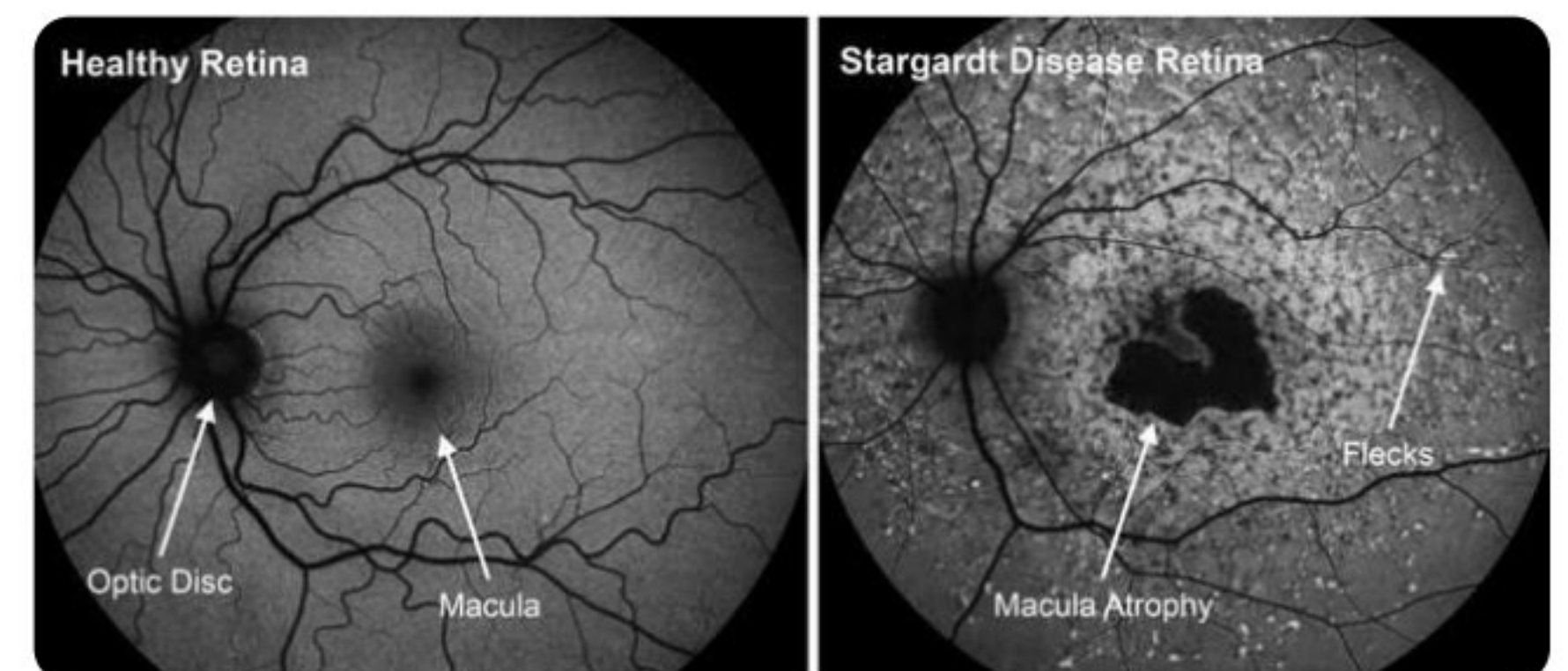
3) Stargardt's disease : affect RPE → (MC) Macular Dystrophy (FUNDUS FLAVIMACULATUS) • AR, chrom 3

- Symptom ↑, sign ↓
- vision loss ? (N) retinal

- sign ⇒ Retinal fleck
 Bull's maculopathy } Bronze beaten macula
 d/t atrophic changes
 in RPE.

In angiography ⇒ DARK choroid

- Inx :: ERG: (N) } @ early stages
 EOG: (N)



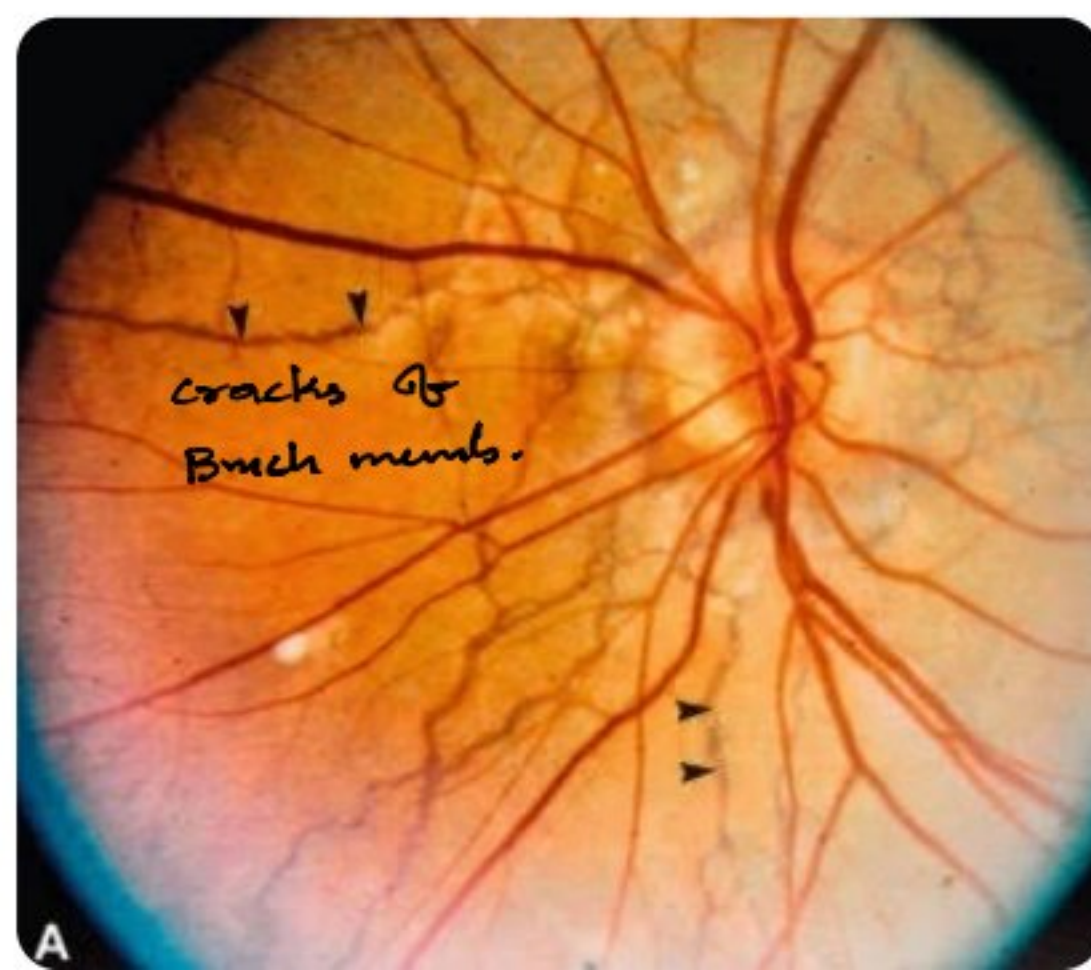
Stargadts d/s - Bronze beaten macula

4) Angiod Streaks → dystrophy of Bruch membrane

causes

- | | | | | |
|-------------------------------------|------------------------------|---------------------|--------------------|---------------------------|
| (P)
psuedo xanthoma
elasticum | (E)
Ehler
Danlos
VI | (P)
Paget
d/s | (S)
Sickle cell | (I)
Incontinentia
P |
|-------------------------------------|------------------------------|---------------------|--------------------|---------------------------|

(MC) → Idiopathic



ERG & EOG pattern of some d/s:

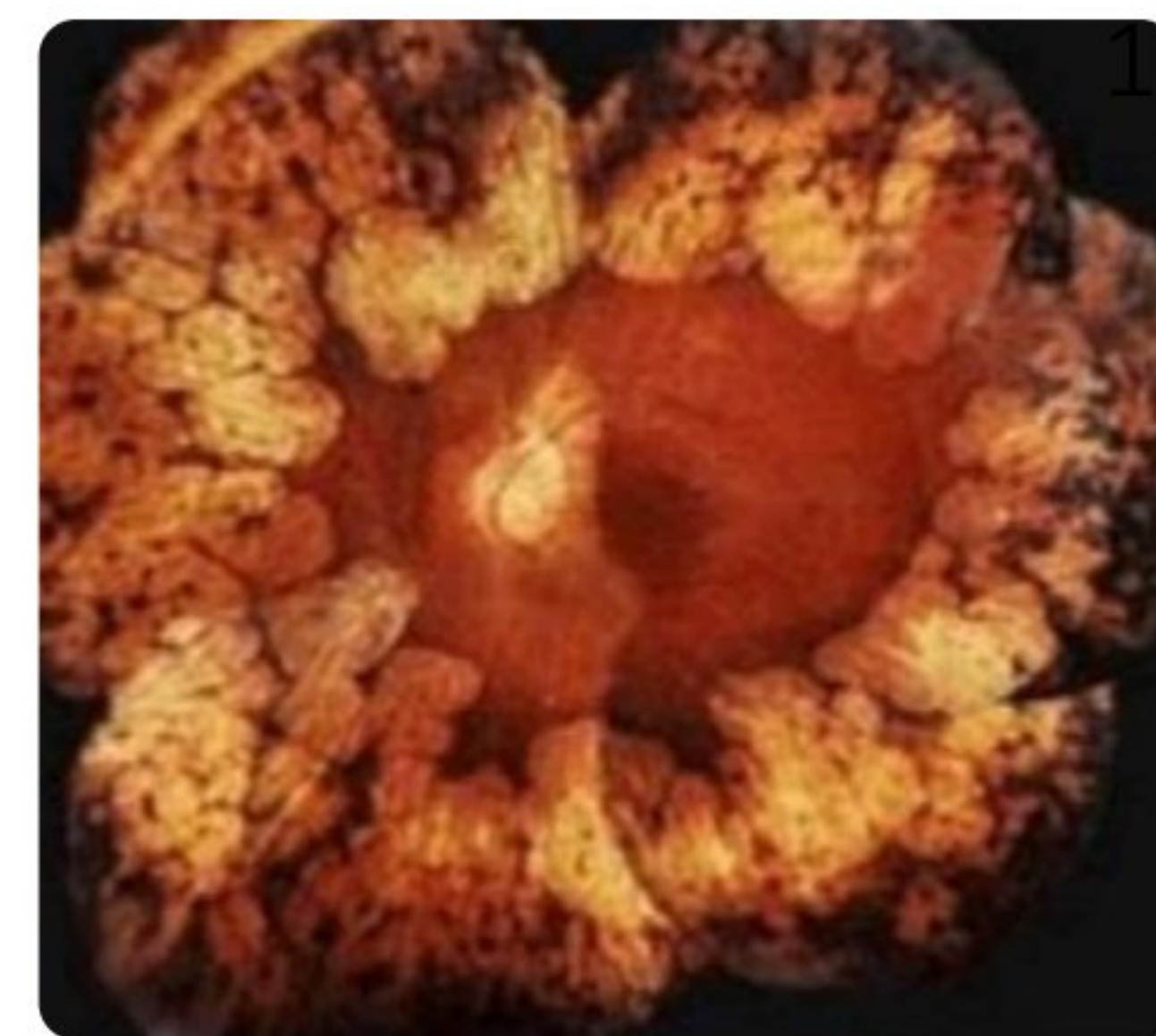
1) RP = ERG = latent a'wave
 EOG = (N)

2) Best d/s : ERG = (N)
 EOG = Ab (N)

3) Stargardt's d/s : ERG = (N)
 EOG = (N)

4) Rod-cone dystrophy : ERG = Ab (N)

- 5) Gyrate dystrophy :: Dystrophy of choroidal vessel
- AR, chromosome 10
 - Ornithine aminotransferase
 - Scalloped areas of RPE loss
 - Ring scotoma
- Rx: Arginine free diet

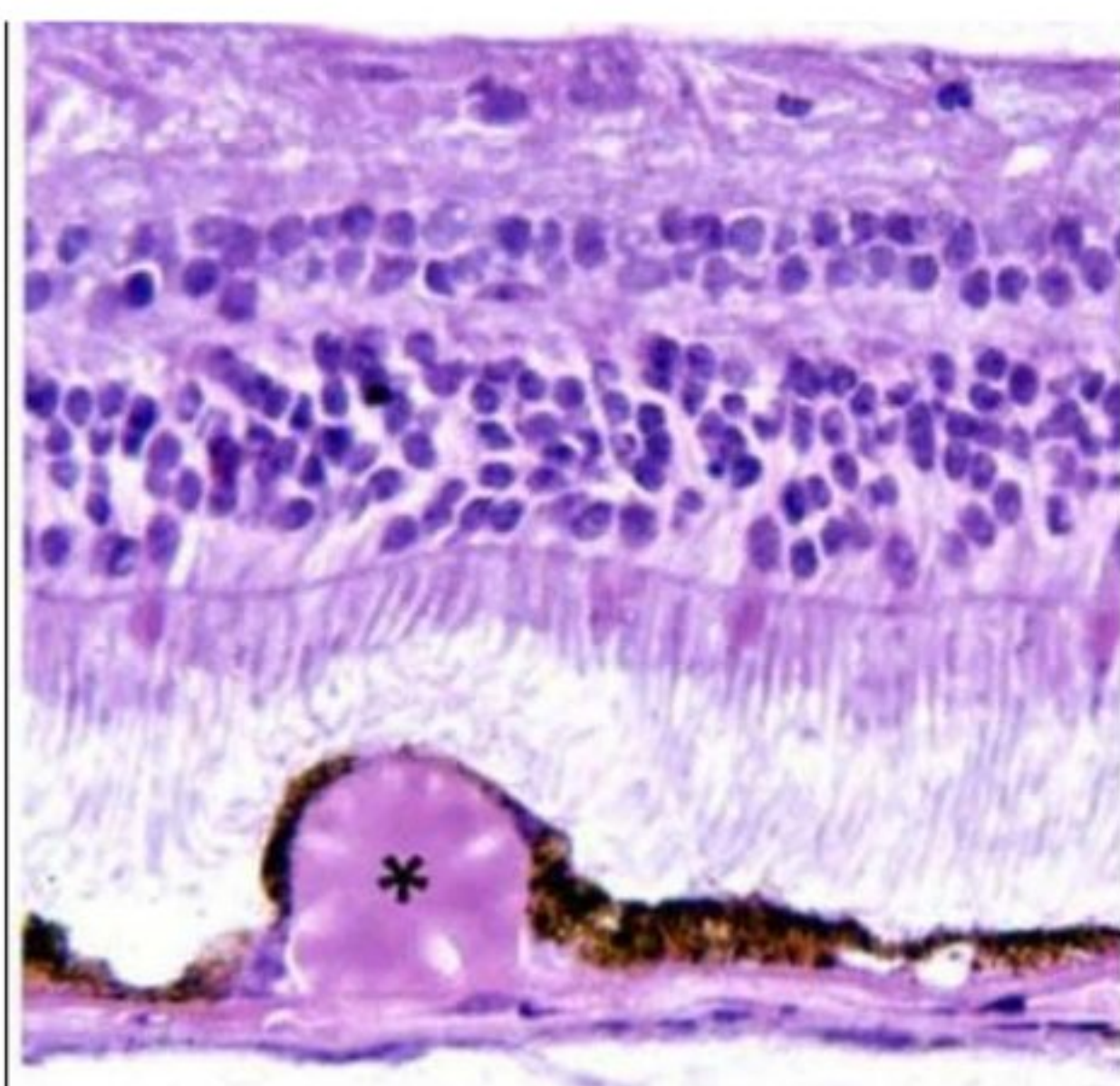


Gyrate Dystrophy

Ring Scotoma

- 1) Pan retinal photocoagulation
- 2) Retinitis Pigmentosa
- 3) Gyrate dystrophy
- 4) Advanced Glaucoma

Q. Identify the pathology in the below histopathology picture of retina.



AIIMS-PG 2018

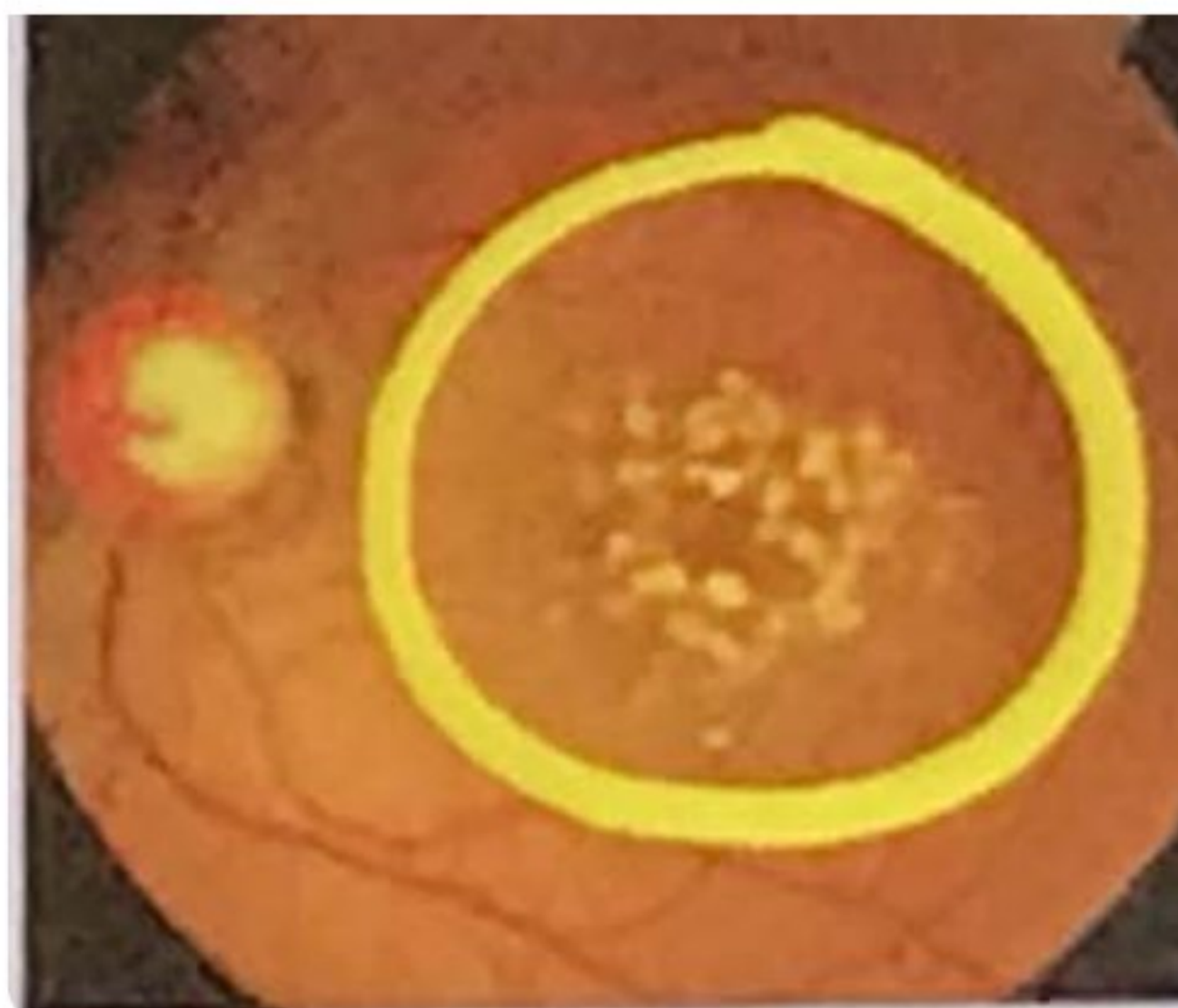
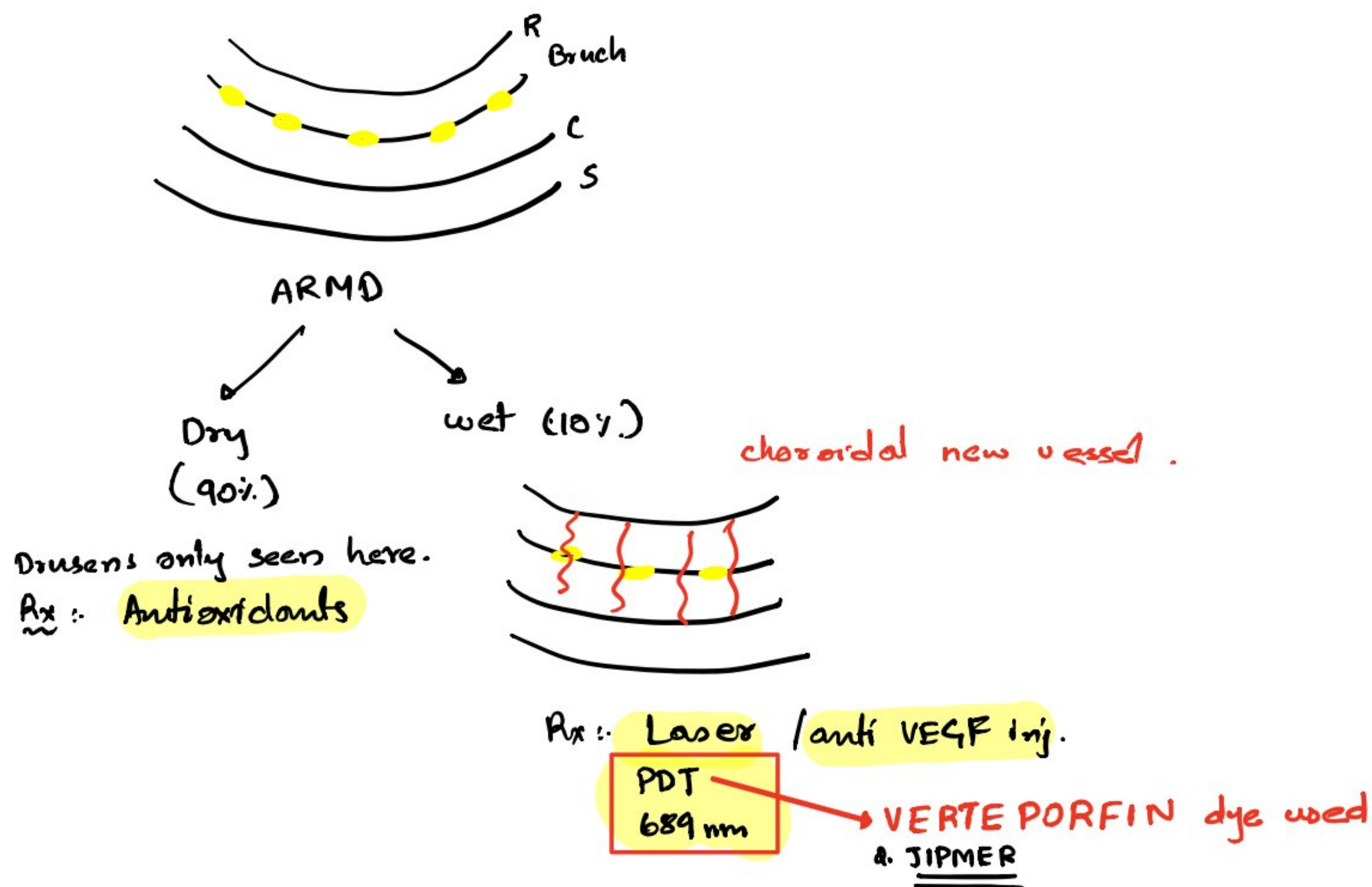
- a) Drusen
- b) Phagosome
- c) Basal linear deposit
- d) Basal laminar deposit

Age-related Macular degeneration (ARMD)

Old Age: Gradual painless vision loss + Metamorphosia

Sign: Drusens in Bruch's membrane

→ Excretory products



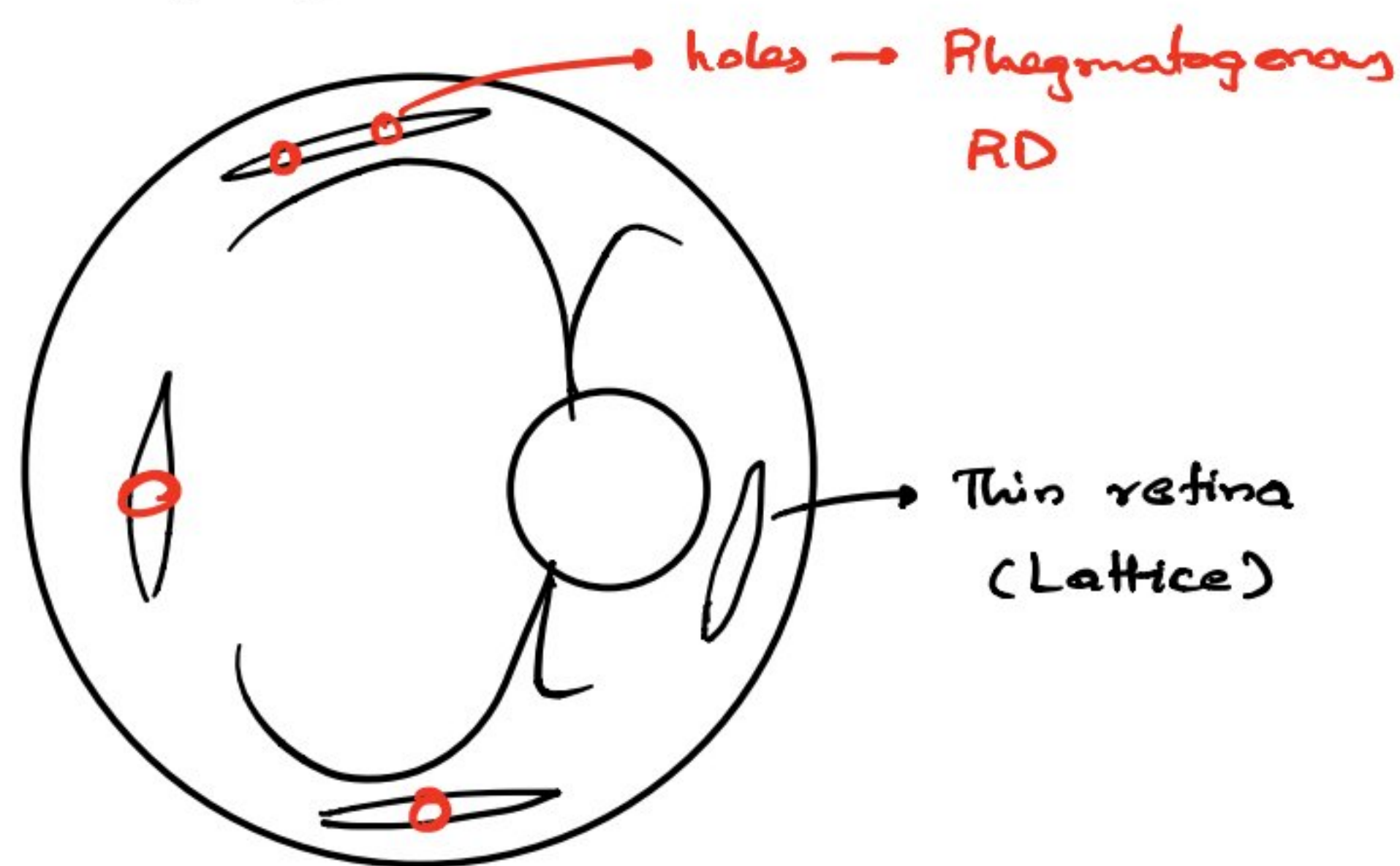
Differentiation b/w H-E & Drusen

Hard Exudates

- 1) Anywhere in Retina
- 2) Accompanied by H'mages
- 3) seen in any age
- 4) Deposited in OPL

Myopic degeneration: Lattice degeneration

- S/I > -6D
- long eye ball

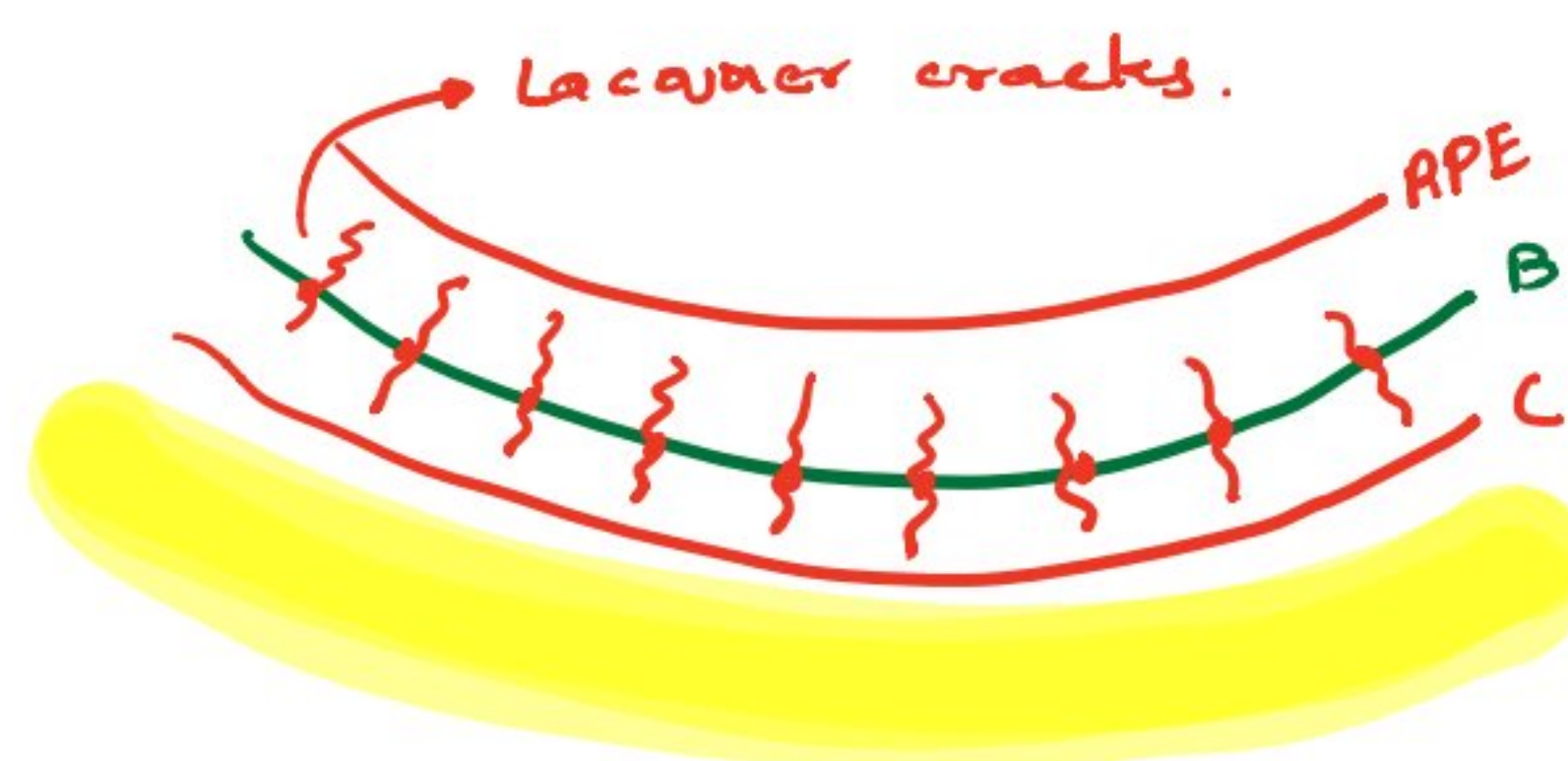


NEET '16 Q. Subretinal H'mage in myopia is K/as.

- a) Lacquer cracks
- ☒ b) Foster fuch spot
- c) Staphyloma
- d) Macular Retinoschisis

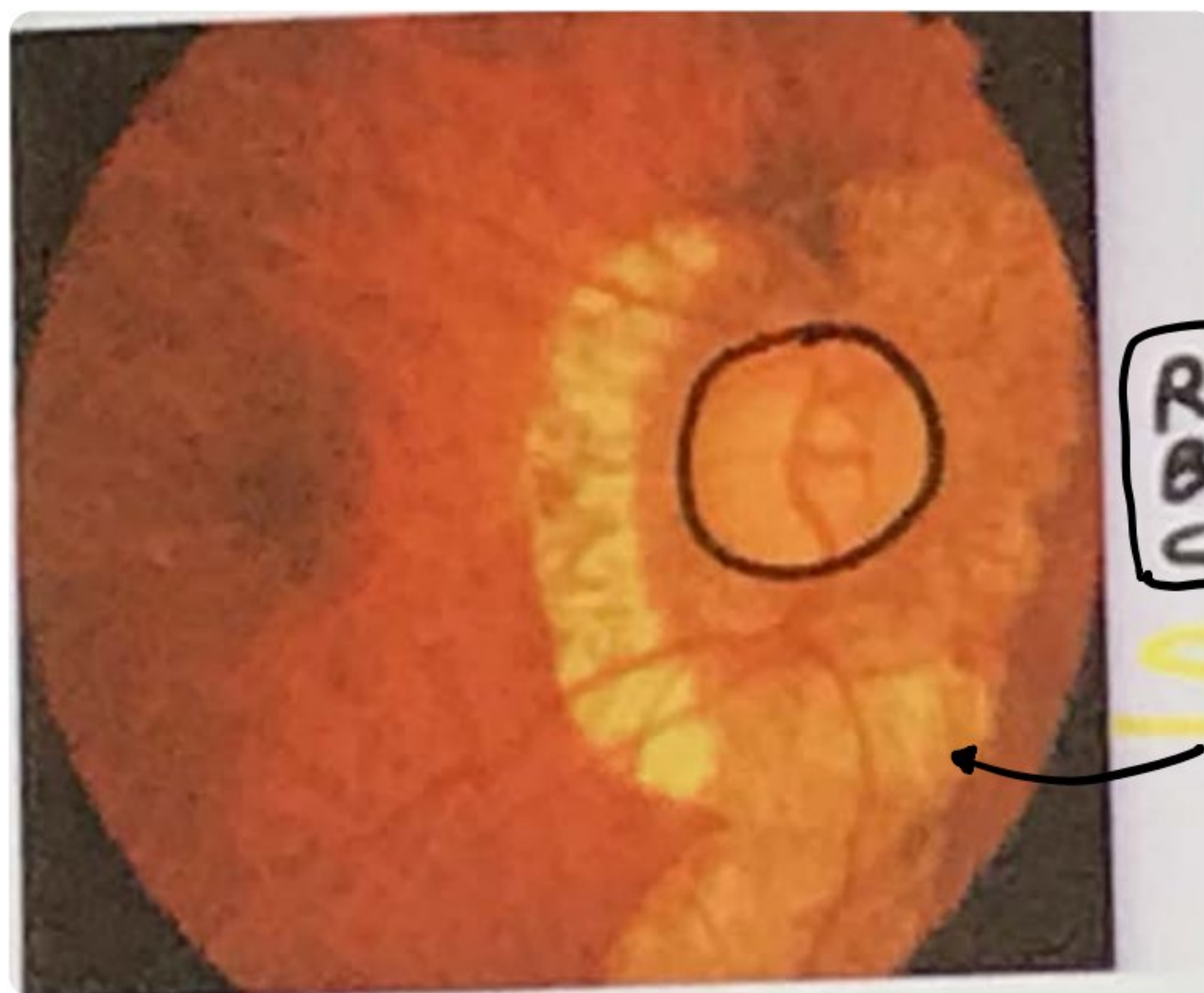
DRUSENS

- 1) In central Retina
- 2) Seen in old age
- 3) Deposited in Bruch



NEET '16

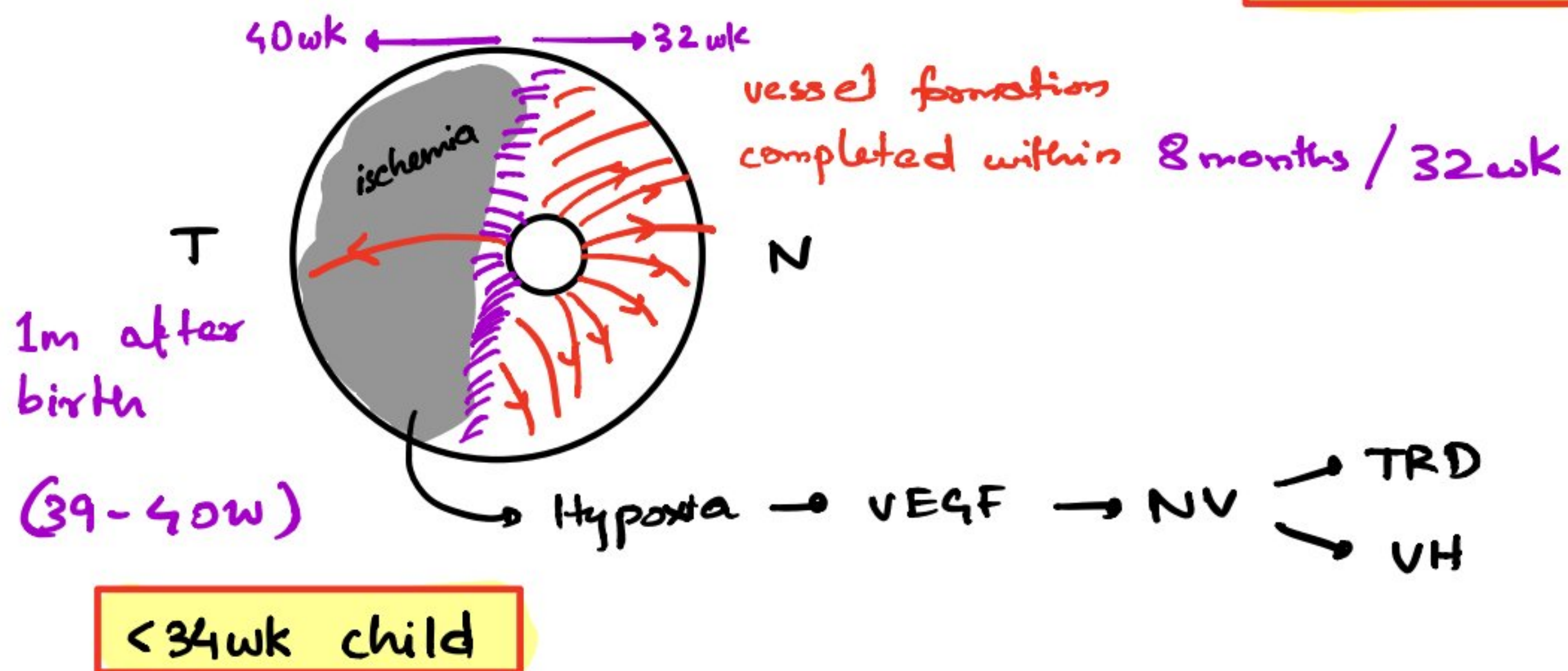
- choroidal NV
- Foster fuch spots ⇒ Remnant of H'mage



↓
Retina
became
so thin
then sclera visible.

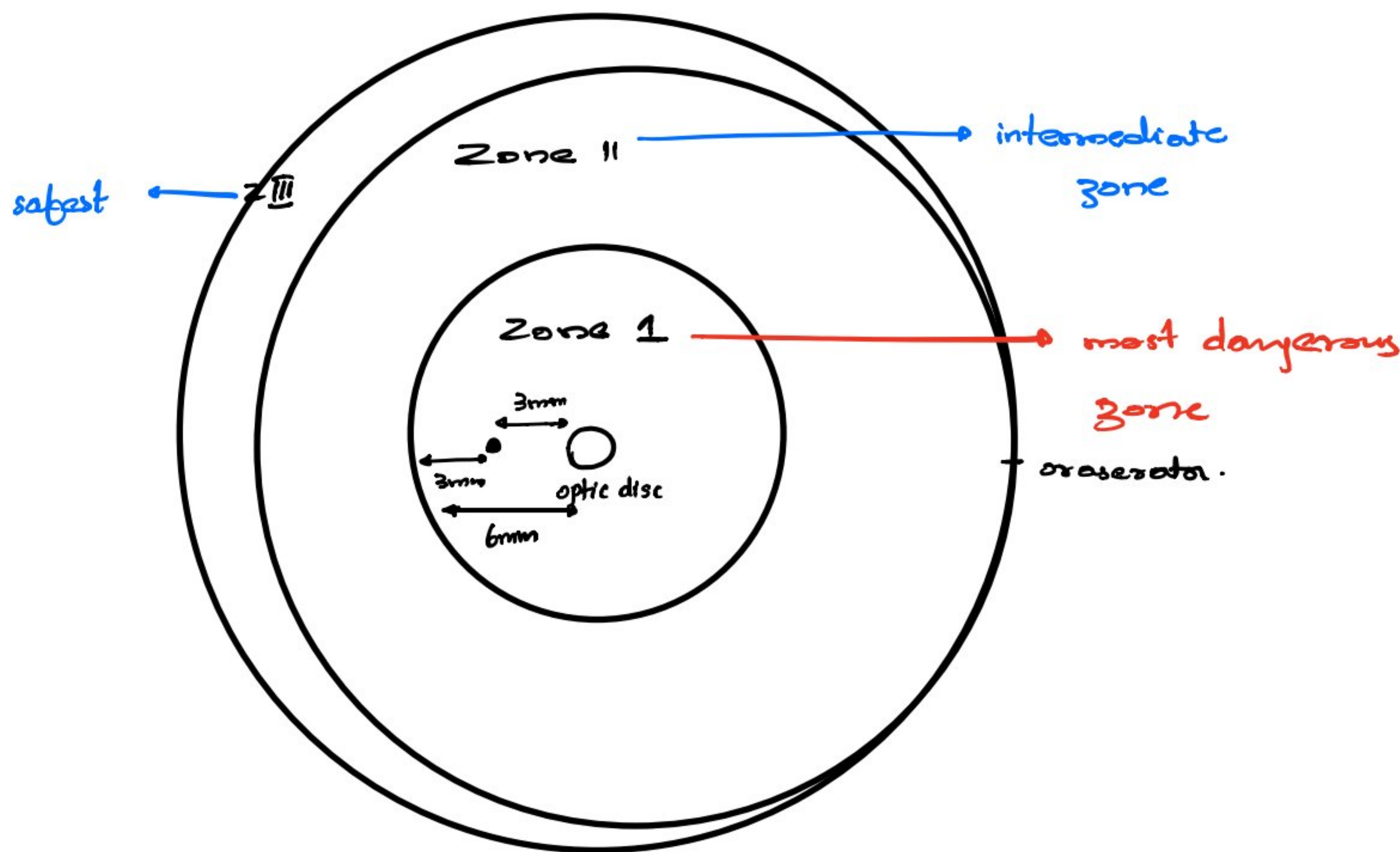
Retinopathy of Prematurity : <34wk

a.k.a ^{APG} RETROLENTAL FIBROPLASIA

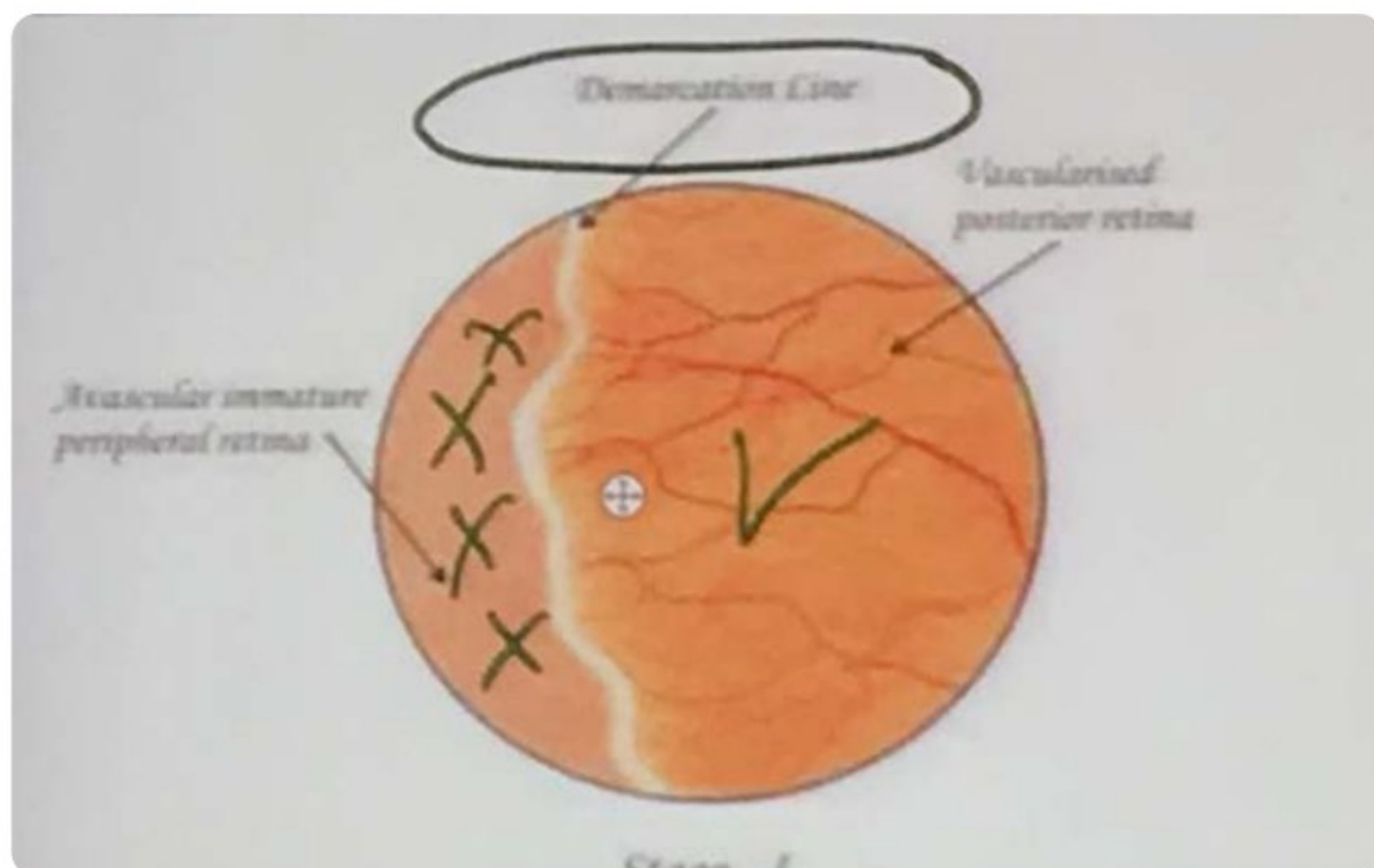


Risk factors:

- 1) <34wk child birth → strongest
- 2) LBW <2000gm
- 3) 100% O₂ (oxygen toxicity) [UPSC]
- 4) Relative Hypoxia



Stage 1: Demarcation line



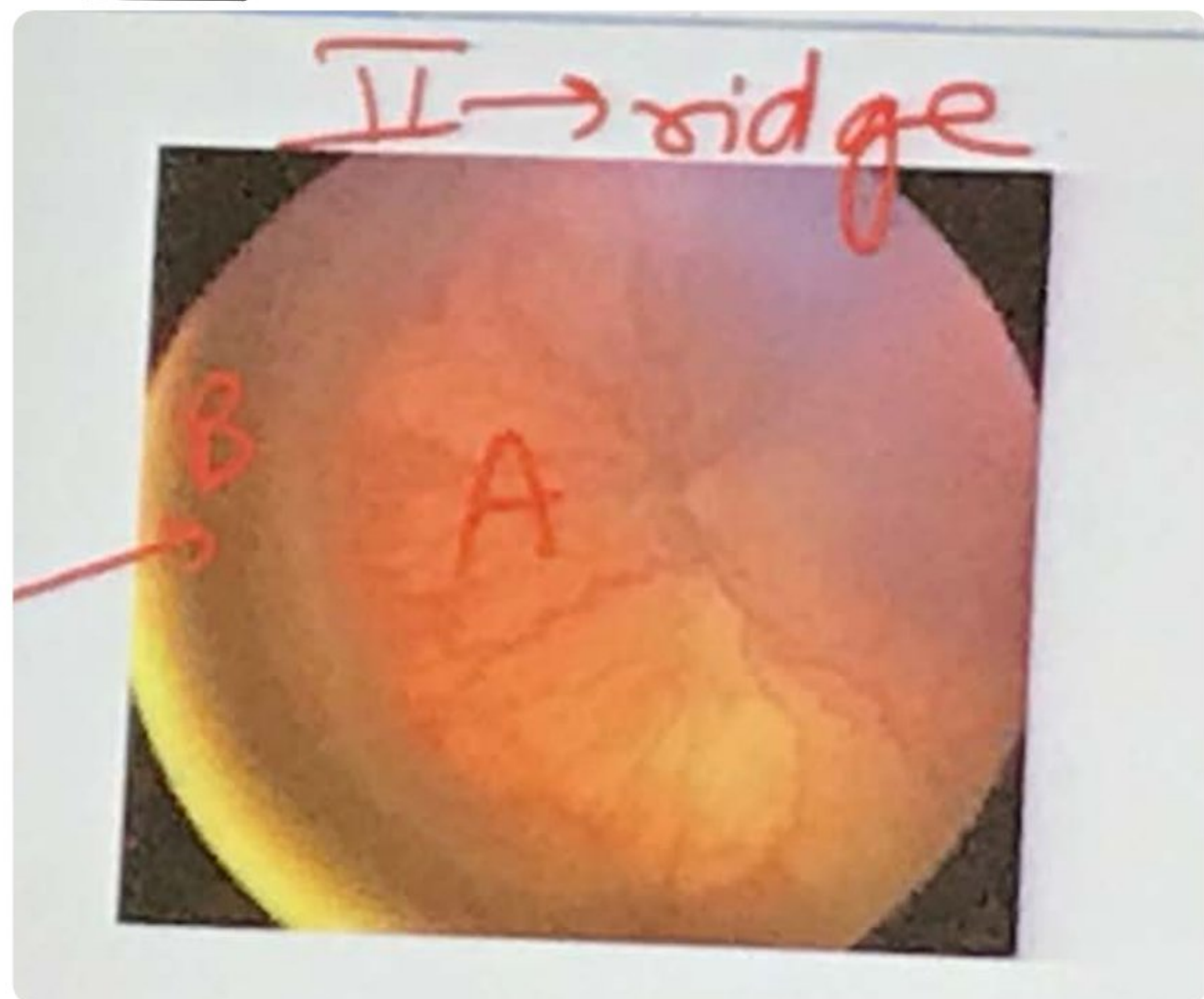
Q: A baby born at 28wk of gestation develop respiratory distress Sx at birth. On day 14, he developed sepsis. After how many weeks should the baby undergo retinal evaluation for retinopathy of prematurity

- a) 4wks
- b) 2wks
- c) 8wks
- d) 12wks

sol. In the scenario above, the postnatal age of the premature infant is 2wks and ∴ the infant must be screened after 2wks.

(i.e 4 week of postnatal age) as per the guideline for the management of retinopathy of prematurity.

Stage II → Ridge formation



NEET PG

Stage III → Fibrovascular Proliferation

Stage IV → Retinal detachment

↓
Partial → (a) → not involving fovea → Rx → Pars Plana Vitrectomy
 → (b) → involving fovea

Stage V → Retinal detachment (TOTAL)

Rx:

1° → Laser the avascular Retina

2° → Anti VEGF treatment

Type 1 ROP:

Zone I + any stage ? ⊕ d/s

Zone I + stage 3 ⊕ out ⊕

Zone II + stage 2/3 ⊕

Non Type 1 is labeled as Type 2 ROP → Zone III → w & w

Screening:

Child born < 28wk
or B.W < 1200g
↓

1st screening @ 3wks of postnatal age

child born ≥ 28wk - 34wk
↓ B.W < 2000g

1st screening @ 4wks of postnatal age

Q: Threshold disease / Threshold retinopathy of prematurity (ROP) is

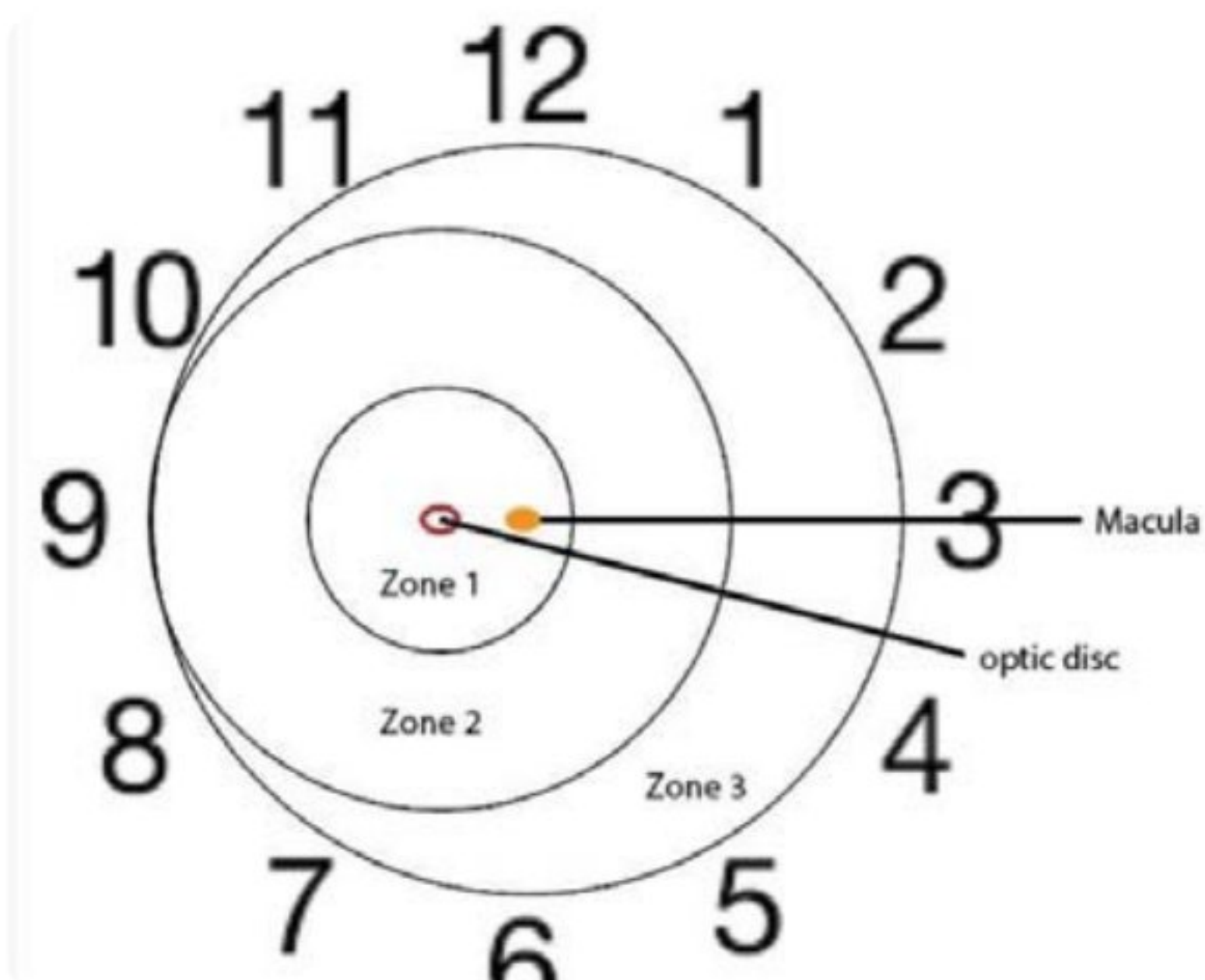
a) Stage 1, Zone 2

b) Stage 3, Zone 1

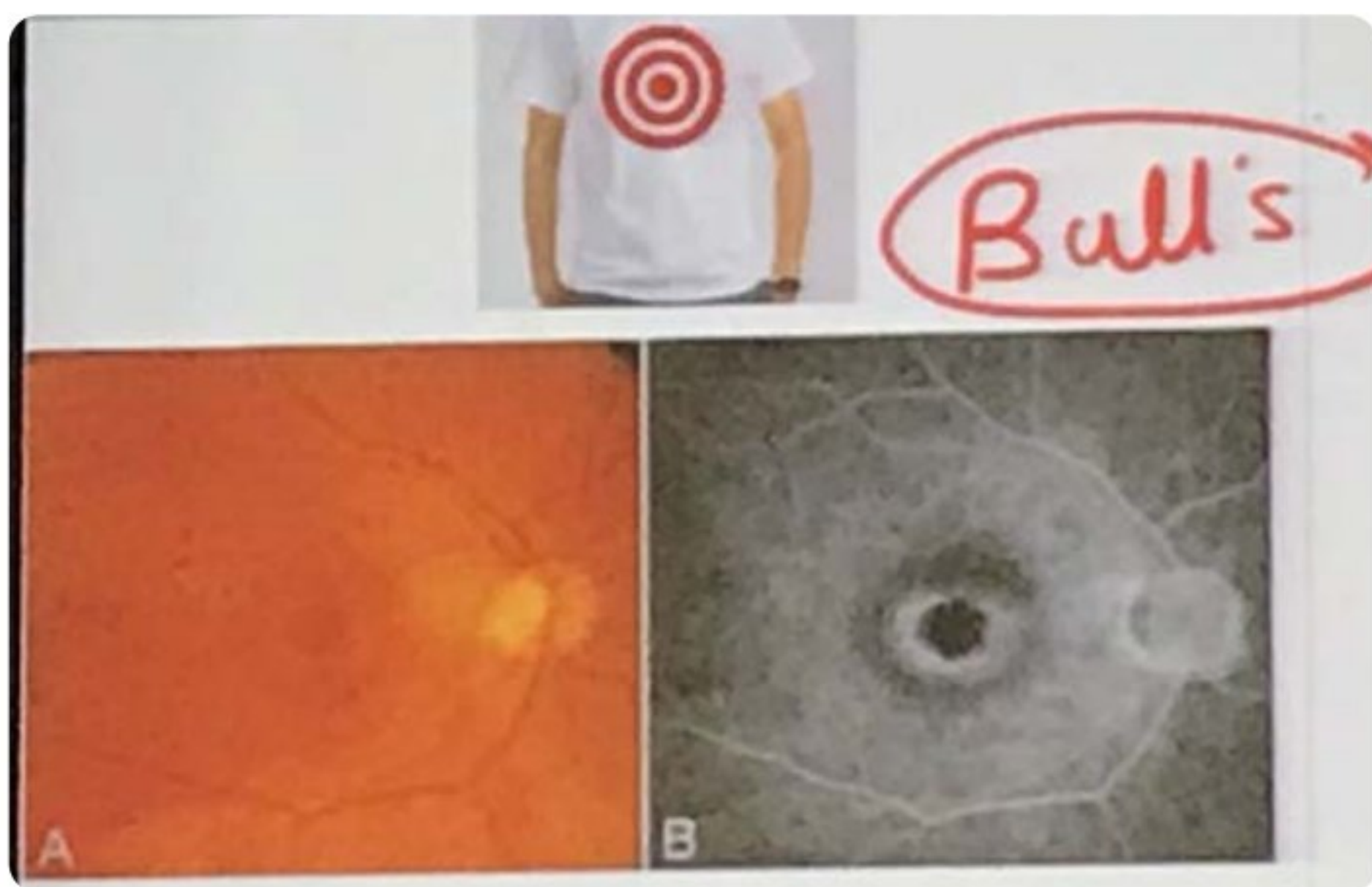
c) Stage 2, Zone 1

d) Stage 3, Zone 3

Treatment for ROP → given only in Threshold retinopathy



Bulls Maculopathy: affects RPE at macular region



Causes ::

- Chloroquine causes Bulls eye maculopathy
- Hydroxychloroquine → for RA patient
- Congenital cone dystrophy
- Juvenile X linked retinoschisis

InxOC: $\neq$ Autofluorescence: appear dark on RPE deficient areas.

Salt & pepper retinopathy

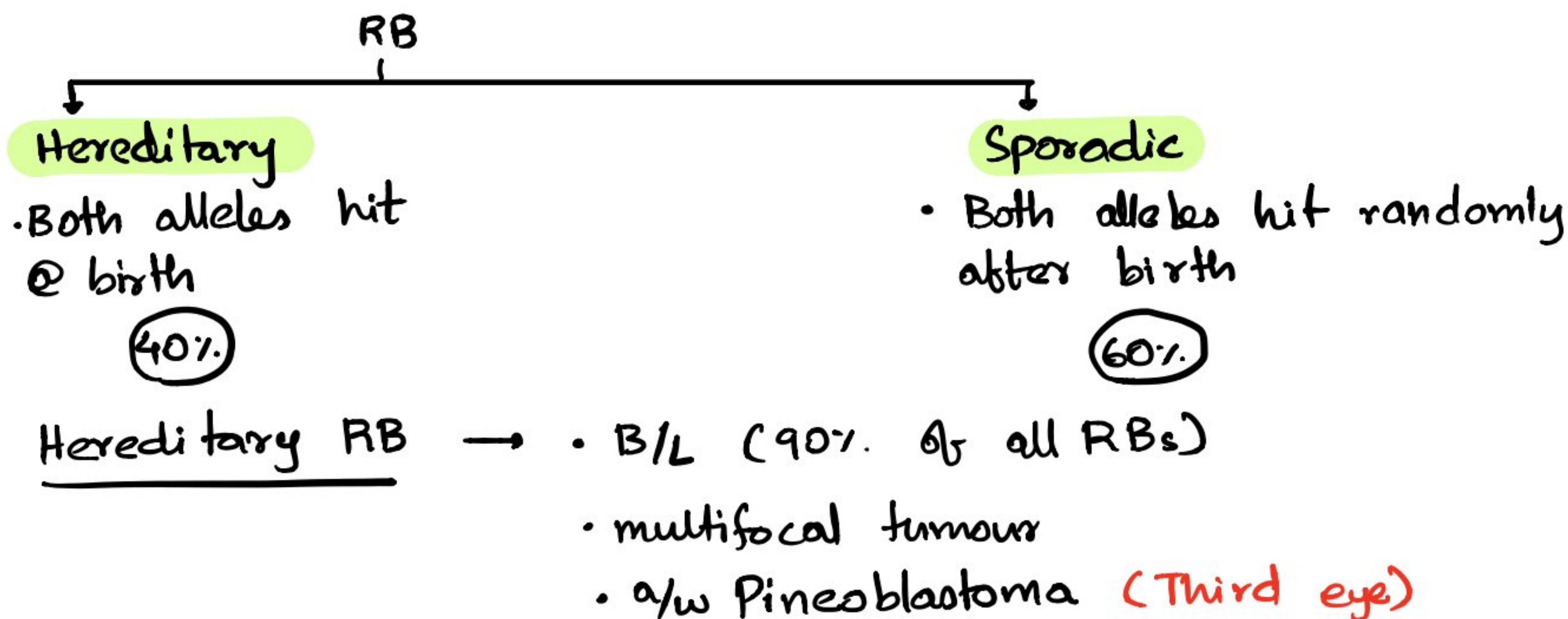
- 1) Retinitis pigmentosa
- 2) Congenital Syphilis → (MCC)
- 3) Phenothiazine toxicity → (MC) drug
- 4) Congenital Rubella
- 5) Old Retinal Detachment
- 6) Quinine.
- 7) Cystinosis
- 8) Leber's congenital amaurosis

Retinoblastoma:

- First ca. to be a/w gene
- RB gene on Chrom 13q14

KNUDSON'S 2 hit hypothesis: One allele already developed a mutation & when 2nd allele develops mutation → RB occurs.

- Avg. age of RB = 18 months
- (MCC) intraocular calcification → RB

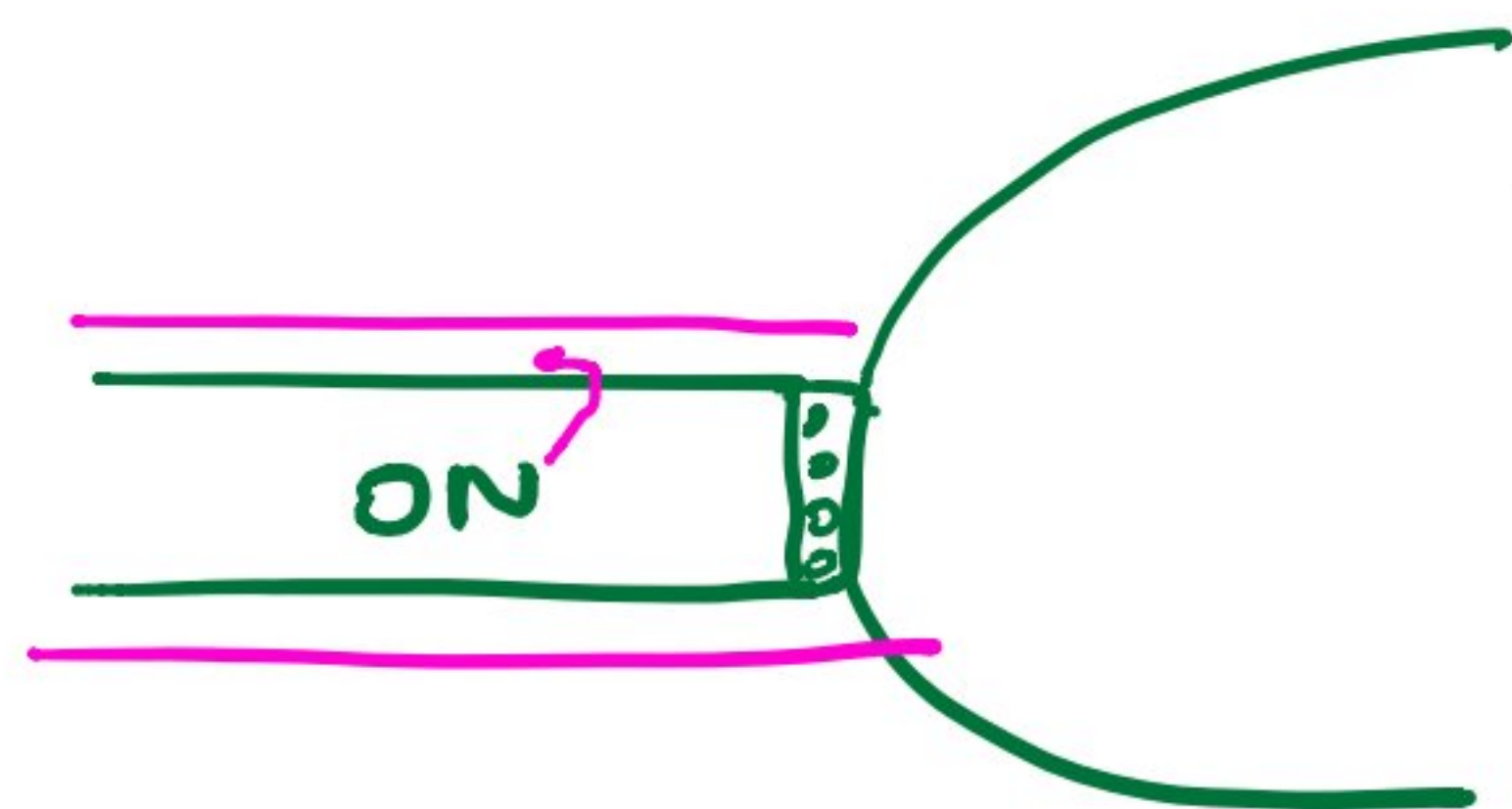


- Trilateral RB = B/L RB + Pineoblastoma
- a/w 2nd malignancy (esp. if RB is treated with radiation)
- carry "AD" germline mutation



50% offspring affected w RB

- RB can spread through CSF



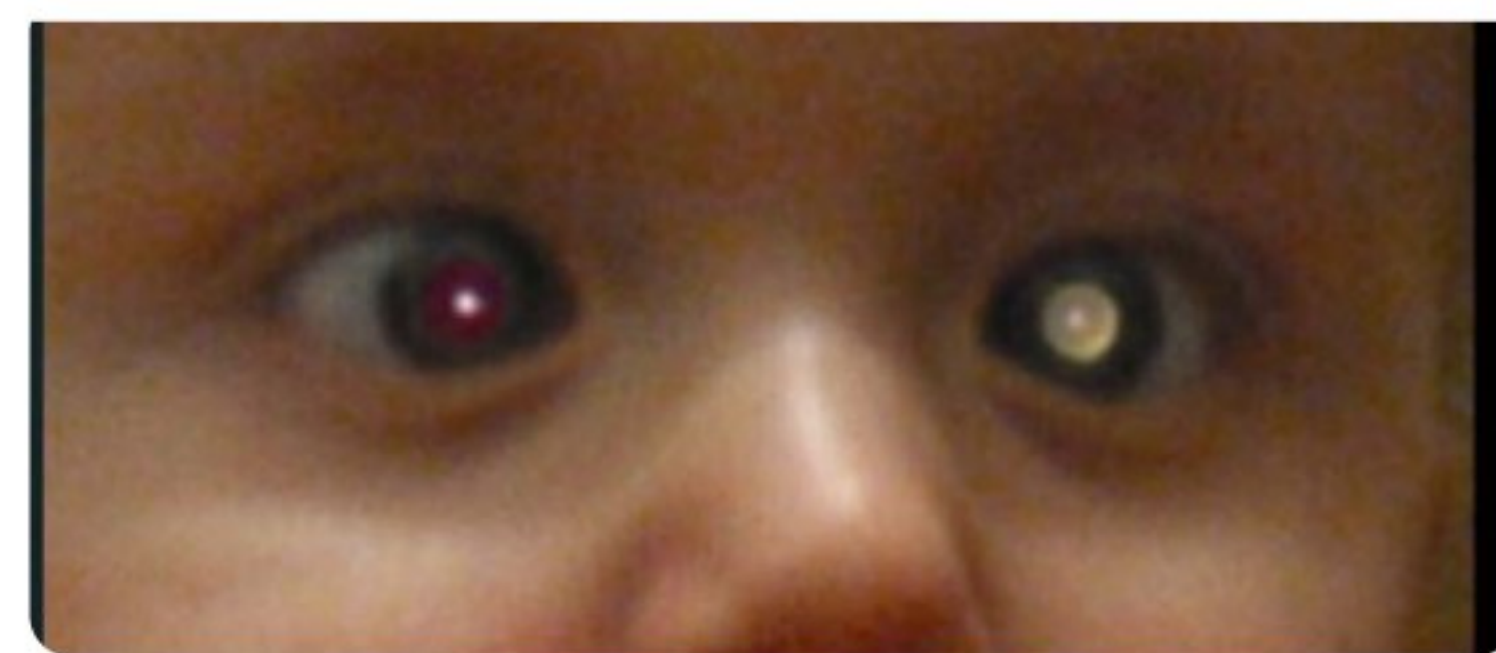
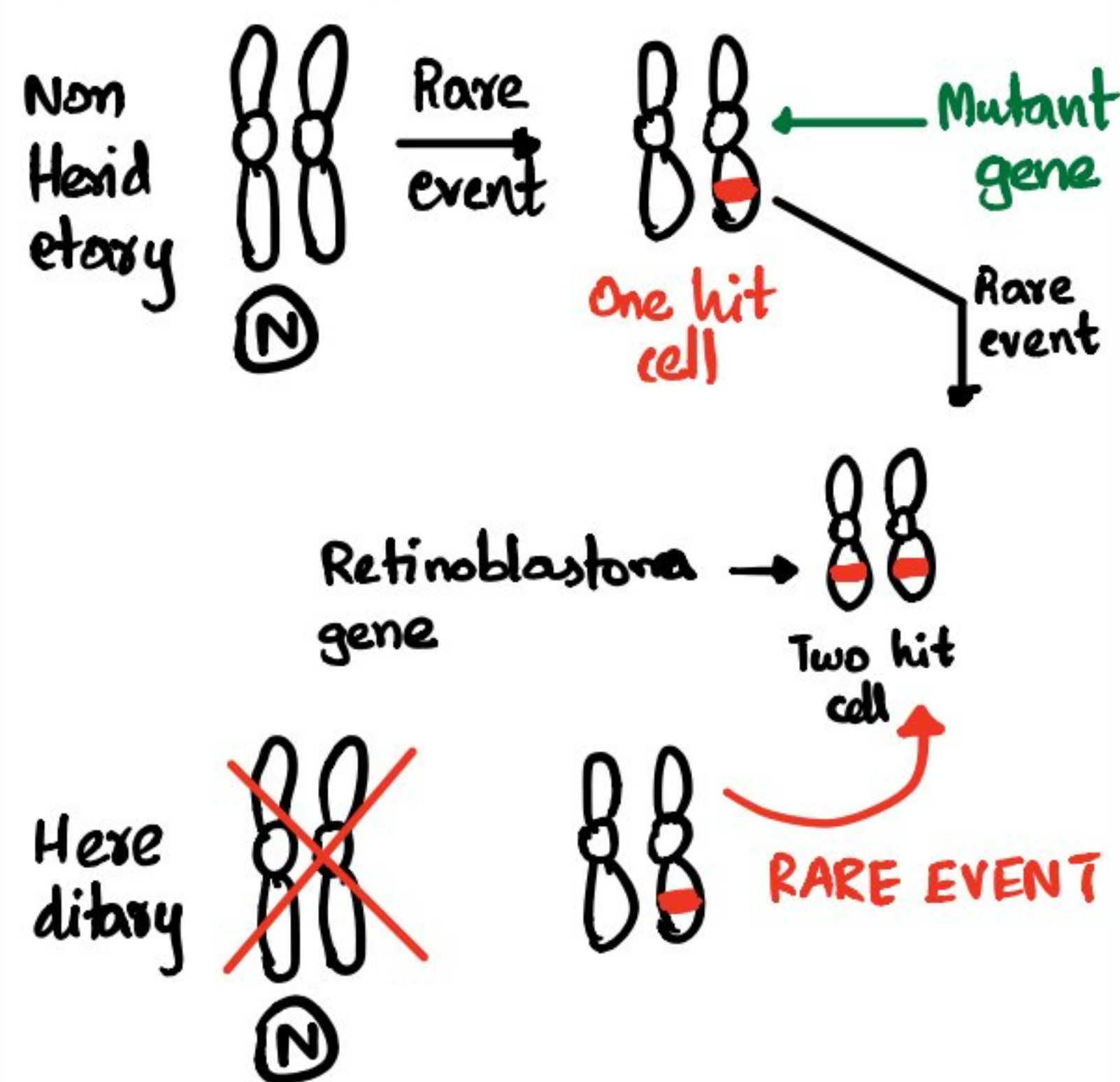
• (MC) site of metastasis = Brain

Imx OC: MRI of orbits + brain
 ↳ for pineoblastoma
 ↳ Brain mets

- C/F:
- Leukocoria (white reflex) → (MC) present
 - Squint (strabismus) → 2nd (MC) present
 ↳ <2yr EUA & fundus examination under anaesthesia
 - Proptosis
 - Orbital Cellulitis
 - Neovascular Glaucoma
 - Phthisis Bulbi [eyeball shrunken]
 - Pseudohypopyon → Tumour cells in AC

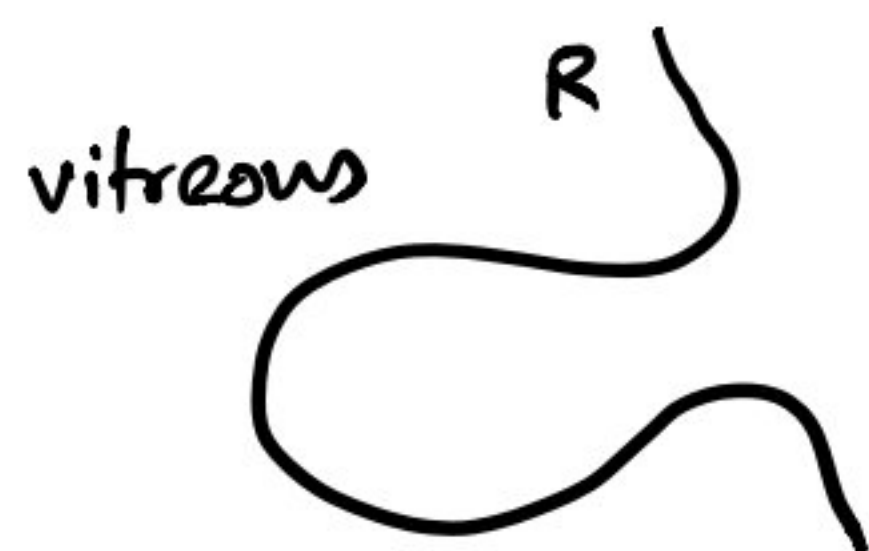
- Inv.:
- USG 'A' scan = **V-Y pattern**
 - Paracentesis: Aqueous humour ↑ sed LDH
FNAC / FNAB → C/I

Two hit theory of Cancer Causation

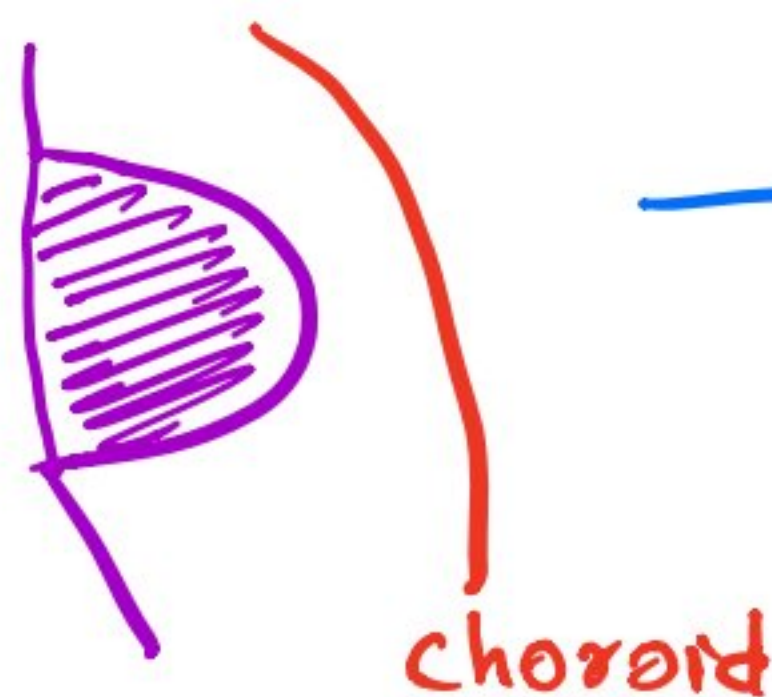


Growth Pattern:

- Endophytic (MC)



- Exophytic



→ Serous RD

- Diffuse Retinal thickening (RETINOXYTOMA)

Rx OC :: Chemotherapy § 1.v§

VEC regimen

Vincristine
Etoposide
Carboplatin } weekly x 6 cycles

Supplementary focal therapy: Laser photocoagulation
↳ Feeding vessels
→ Tumour itself

~~a~~ cryotherapy → [only for peripheral tumours]
(ora serata)

Radiation § If chemo unresponsive§

i) Brachytherapy

ii) Ext. Beam R.T & Proton beam

↓ s/e Hypoplasia of facial bone (DYSMORPHIA)

REESE - ELLSWORTH CLASSIFICATION for RB

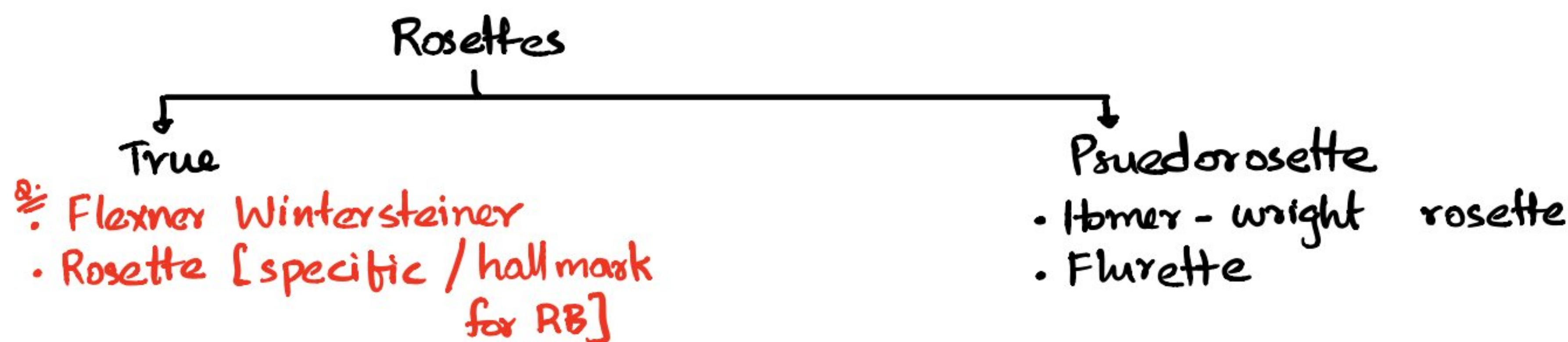
Enucleation: § final resort, if no responding to Radiation§

- Indications → Advanced RB like NVG
Tumour filling $> \frac{1}{2}$ of vitreous

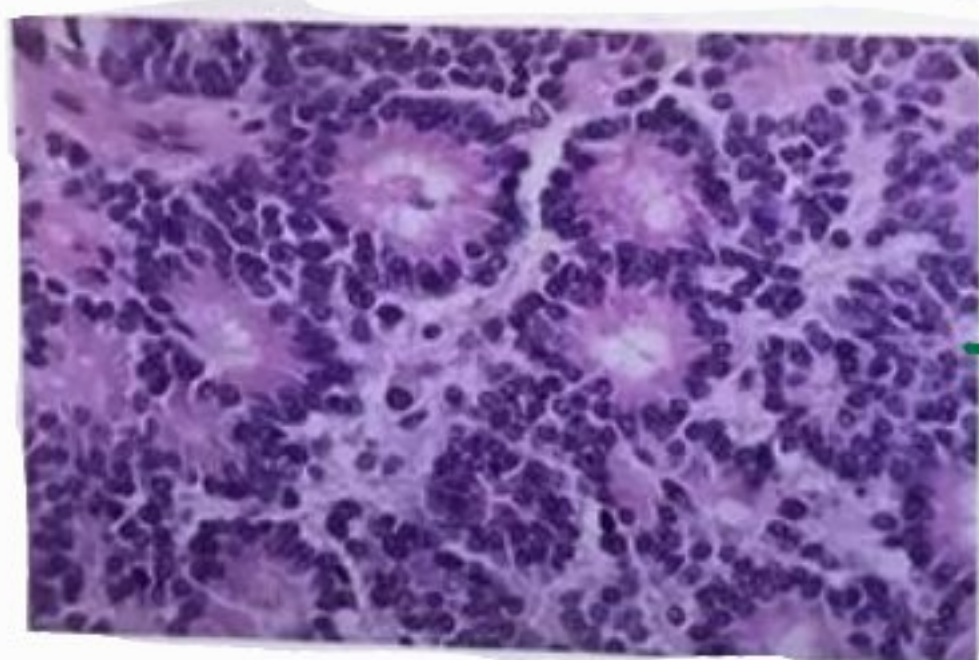
Intra arterial chemo: Direct ophthalmic cannulation [Targeted delivery]

↳ VEC ± Melphalan

Enucleation: → HPE



FLEXNER - WINSTEINER ROSETTE



→ Very well differentiated
Good Prognosis



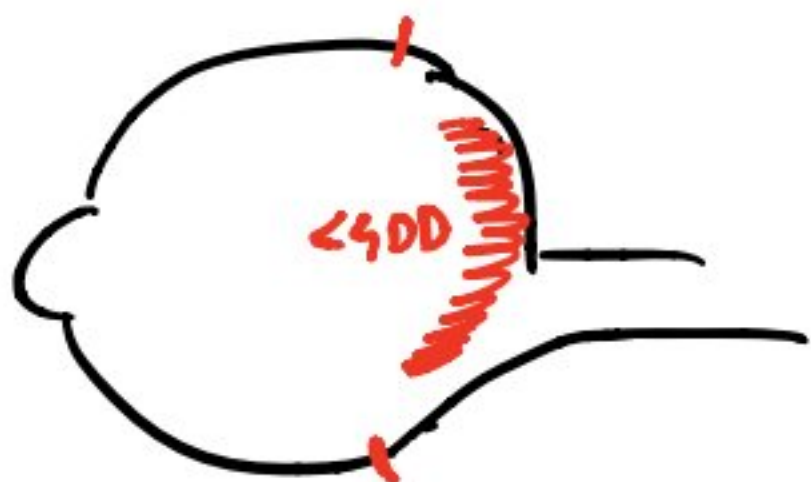
Gross Pathology of RB

↓
Cottage cheese app. of tumour

REESE ELLSWORTH CLASSIFICATION:

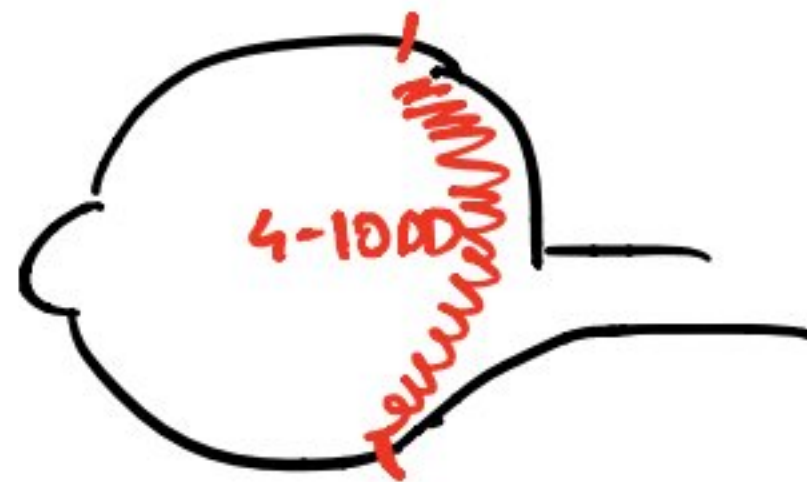
1: Very favourable

Rx: Chemotherapy



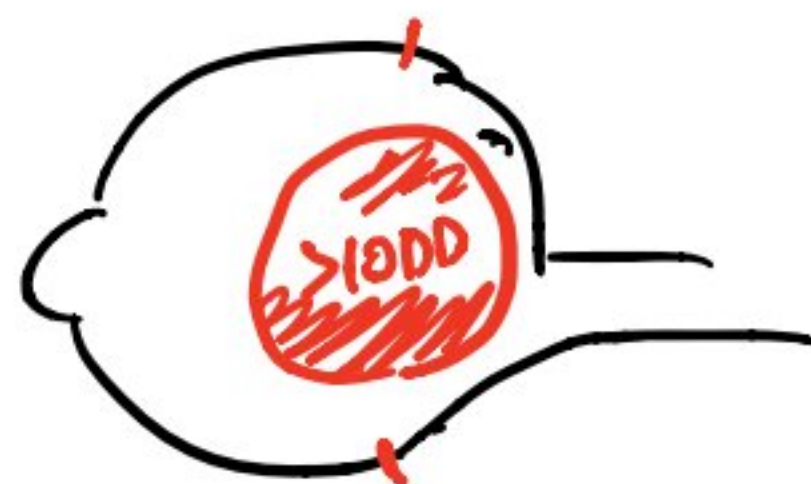
2 Favourable

Rx: Chemotherapy



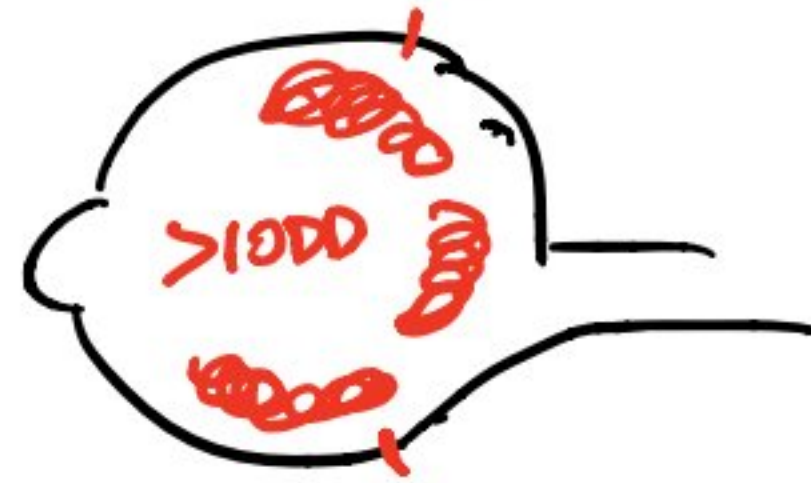
3 Possible

Tx: Cryotherapy



4 Unfavourable

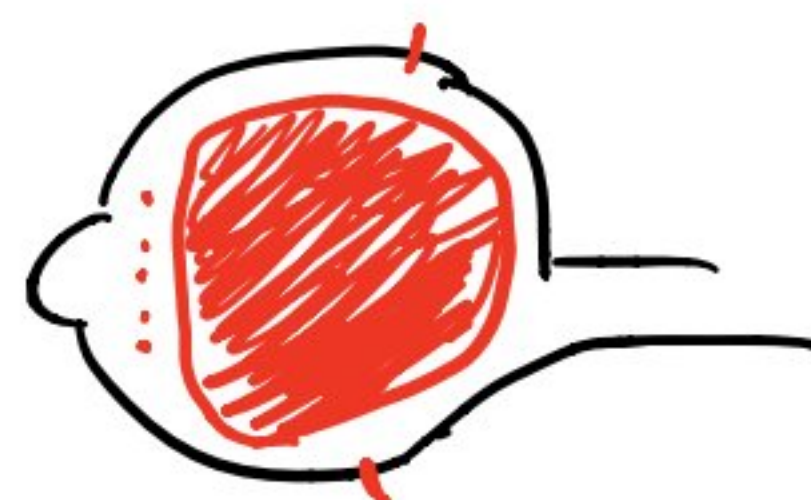
Cryotherapy



5 Highly unfavourable

Rx: Enucleation

+ Postoperative Chemo



Q: Retinoblastoma arises from

a) Pigmentary epithelium

☒ b) Neurosensory retina

c) Optic Nerve

d) None of the above

sol. Retinoblastoma arises from neurosensory retina.

Q: Reese - Ellsworth classification for retinoblastoma is important for _____

a) Management of the tumor

b) Management of the complications

☒ c) Assessment of the survival of the eye after local therapy

d) Assessment of the survival of the pt. after surgery.

Q: What is the treatment of choice for anterior lesions of retinoblastoma?

a) Cryotherapy

b) Photocoagulation

c) Both a and b

d) None of the above

Cryotherapy → used for anterior lesions of retinoblastoma.

Photocoagulation → posterior lesion of retinoblastoma.

Q: The treatment of choice when retinoblastoma involves the Optic N. is _____?

a) Enucleation

b) Enucleation + Postoperative radiotherapy

c) Enucleation + Postoperative radiotherapy + Postoperative chemotherapy

☒ d) Enucleation + Postoperative chemotherapy

Q: The spread of malignant melanoma of the uvea outside at location other than the orbit is treated with _____?

- a) Enucleation
- b) Exenteration
- ☒ c) Enucleation + Chemotherapy + Immunotherapy
- d) Exenteration + Radiotherapy

Q: Which of the following conditions is not termed as Pseudoglioma?

- a) Toxocara Infestation
- b) Norrie disease
- c) Persistent hyperplastic primary vitreous
- ☒ d) Retinoblastoma

Pseudoglioma: is a term given to conditions with signs similar to retinoblastoma. hence, retinoblastoma cannot be called as pseudoglioma.

All other condition listed i.e. **TOXOCARA infestation**

- Norrie disease
- Persistent hyperplastic primary vitreous

Intraocular Pressure (IOP): raised in retinoblastoma and lowered in pseudoglioma.

CHOROID TUMOURS:

Q: Patient presented with painless decrease in vision and metamorphopsia. Fundus examination revealed the lesion shown in the image. What is the expected finding on angiography?

- a) Smoke - stack appearance
- b) Ink blot appearance
- c) Collar stud appearance
- ☒ d) Double circulation (10%)



Q: Which among the following statement is false regarding secondary carcinoma of the choroid?

- ☒ a) Surgical excision of eye is indicated
- b) Tumors are radiosensitive
- c) Patient presents with diminution of vision
- d) None of the above

Picture of a Choroidal Melanoma with Retinal Detachment

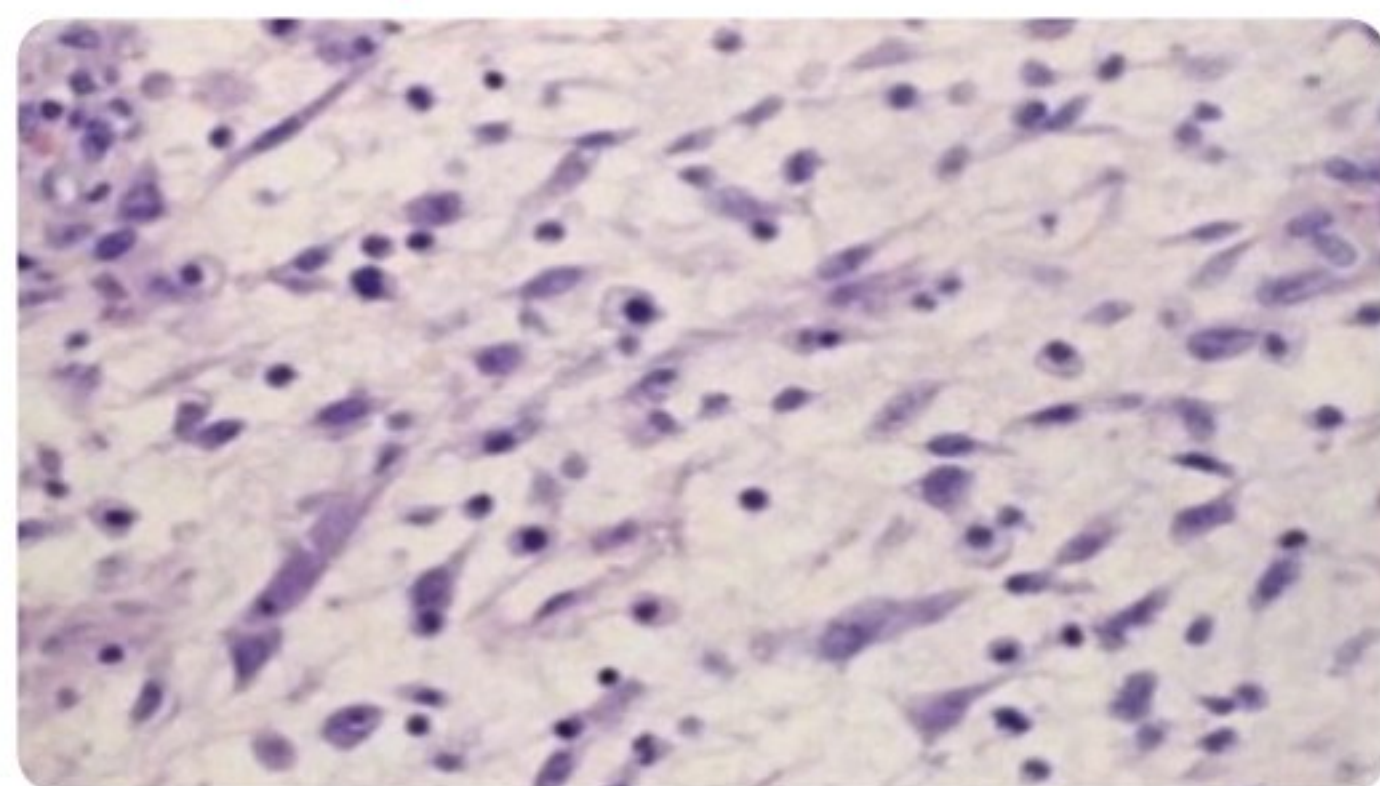


Q. The most common secondaries in choroid are from _____

- a) Breast
- b) Brain
- c) Ovary/Testes
- d) Lungs

RHABDOMYOSARCOMA → (MC) 1° intraorbital tumour in children
(MC) 1° " " in adults → Diffuse large B cell NHL

Q. An 8yr old child presented with right-sided proptosis. Biopsy of the orbital mass was done and histopathological examination revealed this picture. What is the most likely diagnosis?



- a) Retinoblastoma
- b) Metastases from neuroblastoma
- c) Rhabdomyosarcoma
- d) Capillary hemangioma

DNB'14:

Q. Cotton wool spots s/i ::

- i) Hypertension
- ii) SLE
- iii) Eale's d/s

NEET '16

Q. Epiretinal membrane s/i a.k.a Macular Pucker

Post. vitreous detachment → form a transparent layer like a scar tissue in front of retina at the macular region

MH-PG:

Q. Pink fundus can be s/i all the following condition except

- a) Papilloedema
- b) Pseudopapilloedema
- c) Papillitis
- d) Retinitis Pigmentosa

→ Hypermetropia

Features	PAPILLODEMA	PAPILLITIS	PSUEDO PAPILLITIS
Laterality	B/L	U/L	U/L/B/L
Pain	Absent	Present	Absent
Vision	Transient blurred	Sudden vision loss	Defective
Media	Clear	Post. vitreous haze	Clear
Disc Color	Red & juicy	Marked Hyperemia	Reddish
Disc margin	Blurred	Blurred	Not well defined
Disc swelling	2-6D	Not > 3D	Dep. on HM
Peripapillary Edema	+	+	-
Retinal Hmoage	marked	less marked	-
Macula Field	MACULAR STAR	MACULAR FAN	No defect
FFA	Vertical oval dye leak	min. dye leak	No leakage

Roths spots

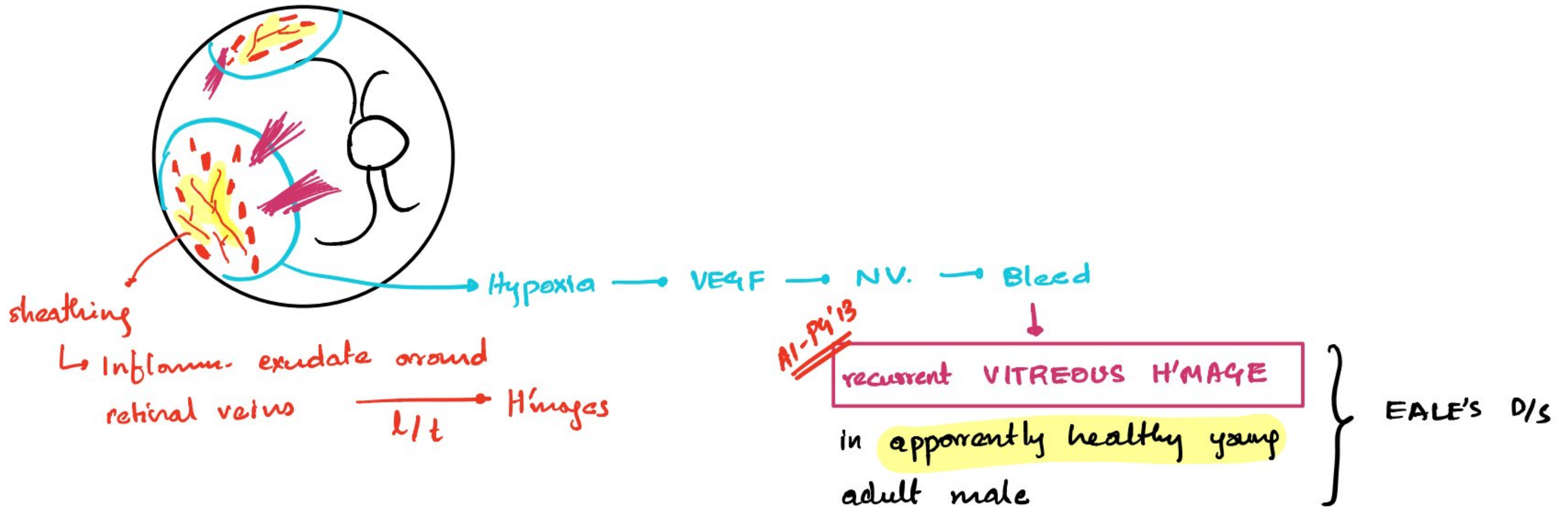


- Retinal hemorrhage with pale centre
- Flame shaped hemorrhage
- Coagulated fibrin, ischemia, inflammation and infection
- Infective endocarditis
- SLE, Leukemia, HIV retinopathy

EALE'S DISEASE

∴ Delayed Type 4 HS to TB protein

• Periphlebitis Retinae → Inflammation of peripheral retinal vein



Management

→ Empirical ATT ⊕ Inflammation
syst. steroids

"

⊕ Retinal hypoxia → VEGF
anti-VEGF intra Vitreous

"

⊕ NV Fluid

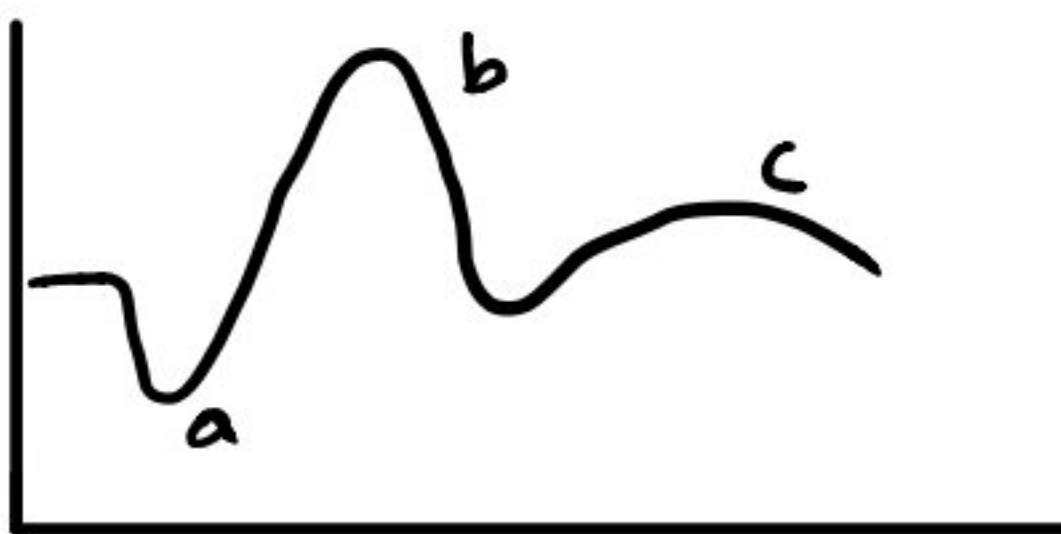
Localised / scatter retinal photocoagulation
complication → VH
→ TRD

"

⊕ Vitrectomy.

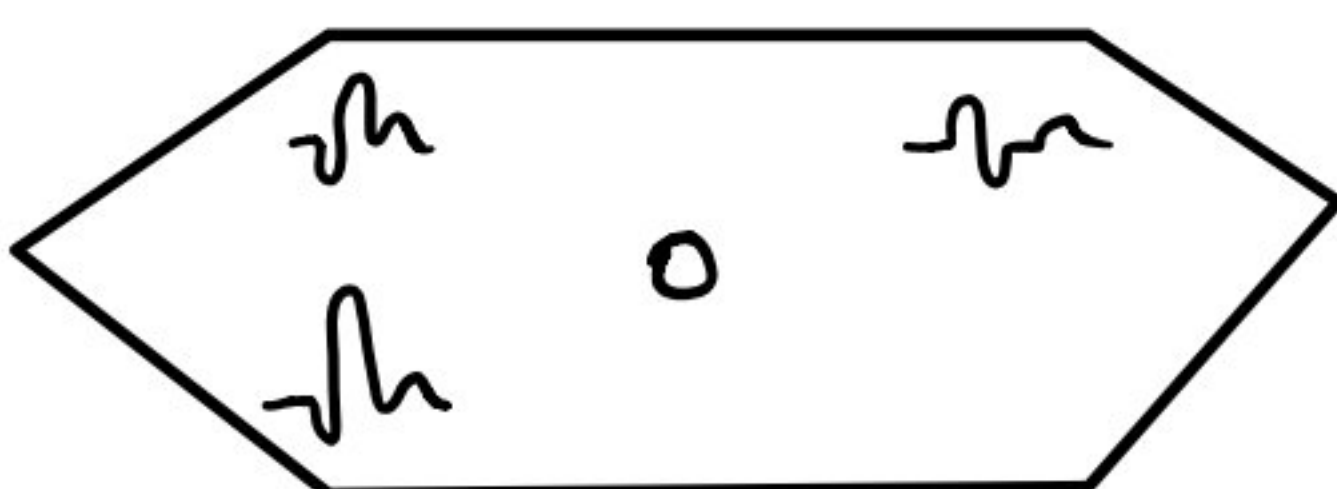
ELECTROPHYSIOLOGY OF RETINA:

1) ERG → Electro retinogram → to assess the neuroophthal component



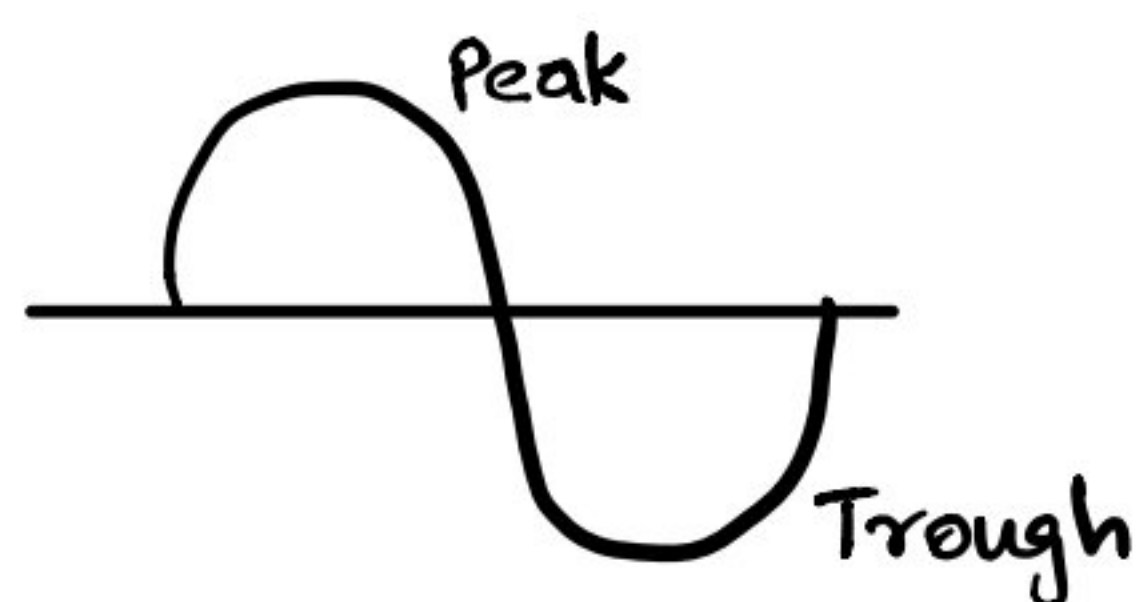
	<u>origin</u>	<u>activity</u>
a →	Photoreceptors	(Rods & Cons)
b →	Muller cells	Bipolar cells
c →	RPE	RPE

2) mf ERG (multifocal)



• Early macular d/o

3) Electrooculogram → Measures the resting potential of Macula ⇒ RPE component.



ARDEN'S Ratio → $\textcircled{N} > 1.85$
Ab $\textcircled{N} < 1.85$

ERG $\textcircled{N}$ } BEST DISEASE
EOG ↓

ERG $\textcircled{N}$ } STARGADT Ds
EOG $\textcircled{N}$

COAT'S D/s:

Q: A 7yr old boy presents with 6/6 vision in the $\textcircled{R}$ eye and vision for only hand movements in the left eye. On fundoscopy, the right eye appears $\textcircled{N}$. The left eye shows retinal detachment, subretinal yellowish exudates, and telangiectatic vessels. What is the most probable diagnosis?

- a) Sympathetic Ophthalmitis
- b) Familial exudative vitreoretinopathy
- c) Retinopathy of prematurity.
- ☒ d) Coats d/s



Ocular manifestation:

- Subretinal exudates
- lipid deposition (hard exudates)
- Vessel may appear telangiectatic with aneurysms that are grape like clustered or light bulb shaped.

SICKLE CELL RETINOPATHY:

Q: What is incorrect regarding sickle cell retinopathy?

- a) Sea fan appearance is s/i supero-temporal quadrant
- ☒ b) HbSS has more severe ocular manifestation HbSC
- c) Angioid streaks are seen
- d) central retinal vein occlusion [CRVO] may be seen.

Hemoglobin SC and S-Thala d/s have the most severe ocular manifestations, although sickle SS d/s has the most severe ocular manifestations, although sickle SS disease has the most severe systemic manifestations.

